26022 – Full Stave Tests & Beamline Tests

Table Runs convention and How to

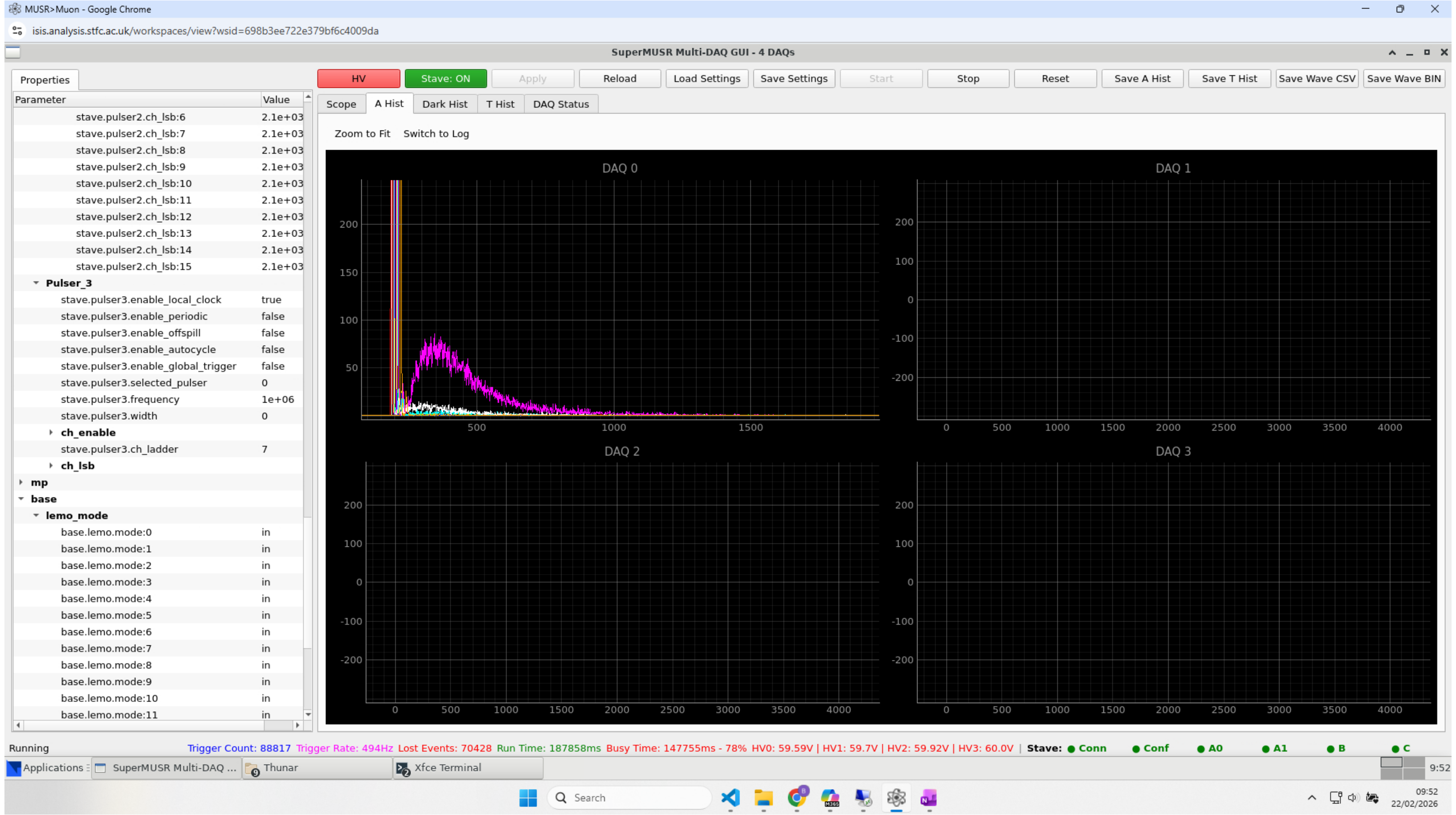
- Give the label run number
- Give the associated “Table Key” a Greek letter identifying the arrangement of tiles, PCB Serial numbers. They can be found on the slides with the tile and PCB arrangements in a table (slides 6 & 7)
- Add notes, including what the run was, the geometery, directory of the data and anything else.

Run Number	Stave Table Key	Notes:
0001	stv_conf_0001	File path on IdaaAS: “ /home/bt1163544\Desktop\beamline\run0001 – 30 30 30 100”. PE is a0:30, a1:30, b:30, c:100. Geometry is focussed on modules b & c. Laying aprox 45 degree on table, 3mm brass degrader
0002	stv_conf_0001	File path on IdaaAS: “ /home/bt1163544\Desktop\beamline\run0002 – 25 25 25 50”. PE is a0:25, a1:25, b:25, c:50. Geometry is focussed on modules b & c. Laying aprox 45 degree on table, 3mm brass degrader
0003	stv_conf_0001	File path on IdaaAS: “ /home/bt1163544\Desktop\beamline\run0003 – 20 20 20 50”. PE is a0:20, a1:20, b:20, c:50. Geometry is focussed on modules a0 & a1. Laying aprox 45 degree on table, 3mm brass degrader, shifted so that positron direction is aligned with a0 & a1.
0004	stv_conf_0001	File path on IdaaAS: “ /home/bt1163544\Desktop\beamline\run0004 – 15 15 20 50”. PE is a0:15, a1:15, b:20, c:50. Geometry is focussed on modules a0 & a1. Laying aprox 45 degree on table, 3mm brass degrader, shifted so that positron direction is aligned with a0 & a1. Bin waves etc saved. Also has a large assortment of RF data.
0005	stv_conf_0001	File path on IdaaAS: “ /home/bt1163544\Desktop\beamline\run0005 – 15 15 20 50”. PE is a0:15, a1:15, b:20, c:50. Geometry is focussed on modules a0 & a1. Laying aprox 45 degree on table, 3mm brass degrader, shifted so that positron direction is aligned with a0 & a1. RF data, focussed on the time histogram. Software threshold set to 600. Cherenkov trigger denoted by “chkv” 22Mhz, “varying_param” means that the RF and LF fields were being varied so, no clear frequencies should be present in the time histogram.
0006		The good RF measurement. We should FFT this to find the 22MHz (think 22.14??). Plus, the ‘beat’ of ~ few us. Dan saved this run. /home/bt1163544/Desktop/beamline/run0006/22MHzRF_Phased_/
---	---	Peter had to remove the detector as they needed to rotate the instrument. No measurements overnights. Andrea analysed the RF data with HV off and made a set of cross talk, FFt pickup, shape is good. We also looked at autocorrelations
0007	stv_conf_0002	File path on IdaaAS: “ /home/bt1163544\Desktop\beamline\run0007 – 15 15 25 50”. PE is a0:15, a1:15 b:25, c:50. Geometry is focussed on modules c. Laying aprox 30 degrees. Binary waves taken. Note this not optimal positioning, no Perspex shim etc. data. Note that offset for ch1 on Module C had an incorrect offset, so this is why the PHS has lower stats.
0008	stv_conf_0002	File path on IdaaAS: “ /home/bt1163544\Desktop\beamline\run0008 – 25 25 35 75”. PE is a0:25, a1:25 b:35, c:75. Geometry is focussed on modules c. Laying aprox 30 degrees. Binary waves taken. Note this not optimal positioning, no Perspex shim etc. data. Note that offset for ch1 on Module C had an incorrect offset, so this is why the PHS has lower stats. High rate on the slits. Very high PE interesting nonlinearity on the time histogram.
0009	stv_conf_0002	File path on IdaaAS: “ /home/bt1163544\Desktop\beamline\run0009 – 15 15 25 50”. PE is a0:15, a1:15 b:25, c:50. Geometry optimised for module B. . Binary waves taken. **Forgot screenshots** ☹

Run Number	**Stave Table Key**	**Notes:**
0010	stv_conf_0002	File path on IdaaAS: " /home/bt1163544\Desktop\beamline\run0010 – 25 25 35 75". PE is a0:25, a1:25 b:35, c:75. Geometry is focussed on modules B . Laying approx. 48 degrees. Binary waves taken.
0011	stv_conf_0002	File path on IdaaAS: " /home/bt1163544\Desktop\beamline\run0011 – 15 15 25 50". PE is a0:15, a1:215, b:25, c:50. Geometry is focussed on modules A0. Excellent Stats Binary Waves Taken
0012	stv_conf_0002	File path on IdaaAS: " /home/bt1163544\Desktop\beamline\run0012 – 25 25 35 75". PE is a0:25, a1:25 b:35, c:75. Geometry is focussed on modules A0. Binary waves taken. DAQ1 Ch1 has poor offset, should be manually changed.
0013	stv_conf_0002	File path on IdaaAS: " /home/bt1163544\Desktop\beamline\run0011 – 15 15 25 50". PE is a0:15, a1:215, b:25, c:50. Geometry is focussed on modules A0. Odd channel behaviour on ch2, seems to have about half of the PHS peak. Tile is to be swapped.
0014	**stv_conf_003**	I didn't save this in-time-order. The name is correct. Same daq config as 0013. but new tile config
0015	**stv_conf_003**	Played with deconvolution - 0015
0016	**stv_conf_003**	Save this as a "long run" of 0014.

Run Number	Stave Table Key	Notes:

Stave PCB: 20718
All 32 channels working 22/02/26 10:13
PHS is working for DAQ0
Light tight except for the corners and junctions between modules – cause is light between metal alignment plate and PCB 22/02/26 10:33
All channels verified to be working and producing photons with the sr90 source 22/02/26/ 10:56

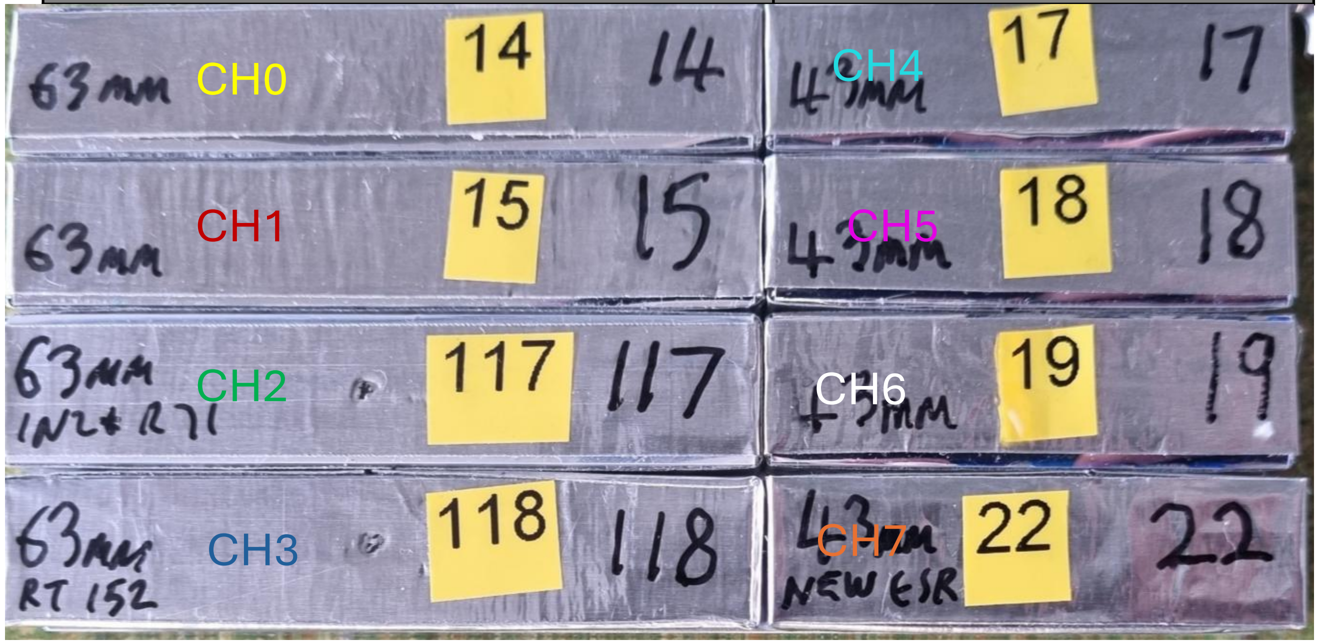


PCB mapping – as looking down on the stave with the shorter tiles on the RHS (Mod B for example)

63mm

43mm

CH0	CH4
CH1	CH5
CH2	CH6
CH3	CH7



Sample will be here in final build

Stave Setups and Tile arrangements 20718

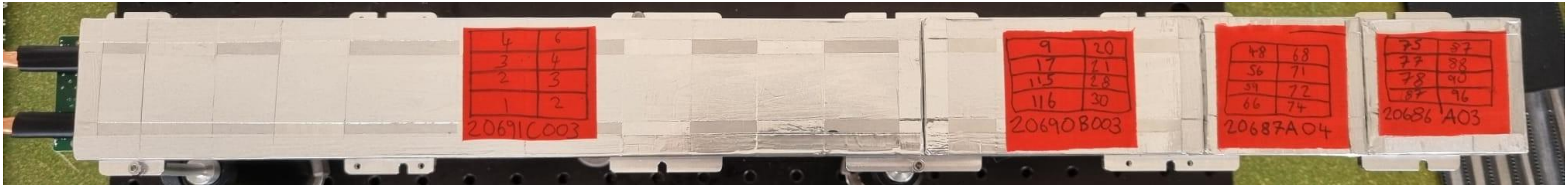
PCB: 20691
Total ID: 20691C003

Note: the image is flipped bc tiles were installed flipped. The sticker (below) is correct.

PCB: 20690
Total ID:
20690B003

PCB:
20687
Total ID:
20687A004

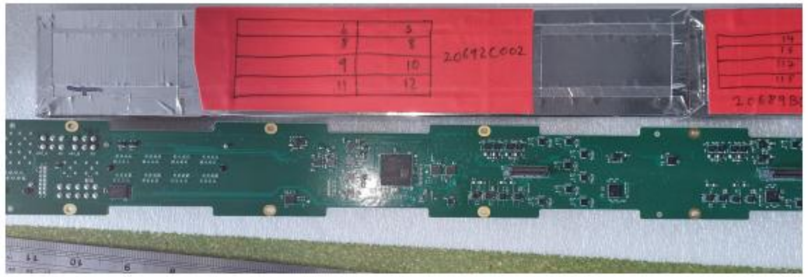
PCB: 20686
Total ID:
20686A003

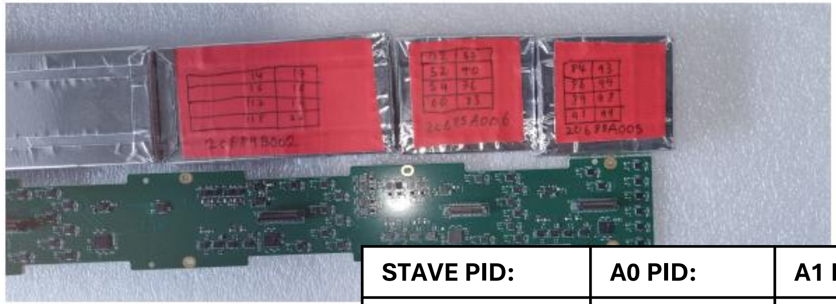


Magnetic Pins

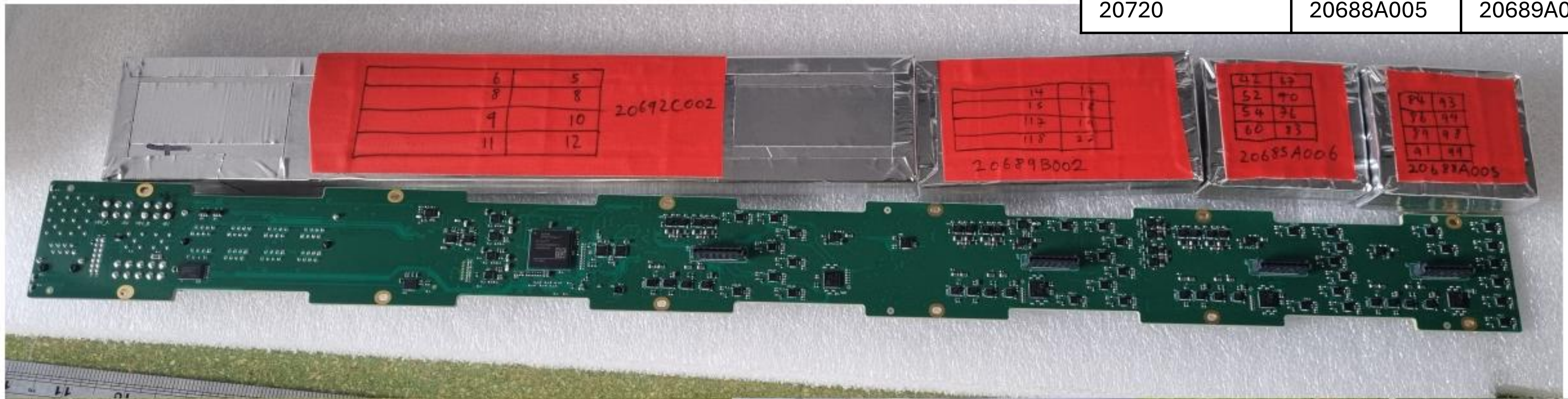
STAVE PID:	A0 PID:	A1 PID:	B PID:	C PID:	Table Key
20718	20686A003	20687A004	20690B003	20691C003	stv_conf_0001

Stave Setups and Tile arrangements 20720





STAVE PID:	A0 PID:	A1 PID:	B PID:	C PID:	Table Key:
20720	20688A005	20689A006	20689B002	20692C002	stave_conf_0002



PCB: 20692
Total ID: 20692C002

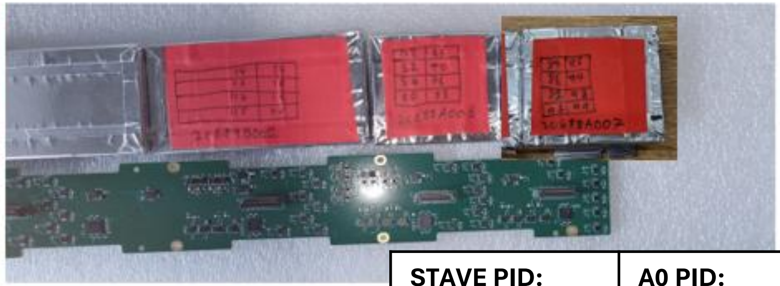
Magnetic Pins

PCB: 20689
Total ID:
20689B002

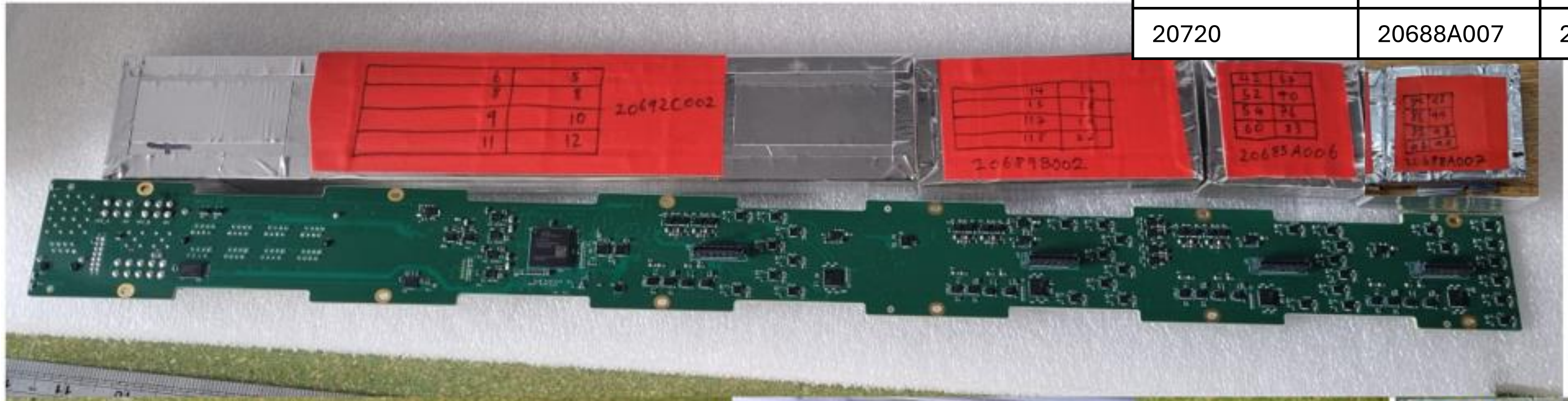
PCB: 20685
Total ID:
20689A006

PCB: 20688
Total ID:
20688A005

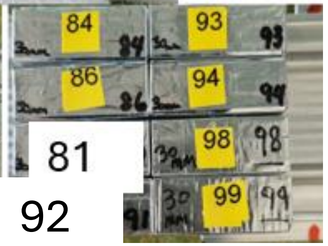
Stave Setups and Tile arrangements 20720 replaced tile



STAVE PID:	A0 PID:	A1 PID:	B PID:	C PID:	Table Key:
20720	20688A007	20689A006	20689B002	20692C002	stave_conf_0003



81, 92, 94,90 are freshly coupled



PCB: 20692
Total ID: 20692C002

PCB: 20689
Total ID: 20689B002

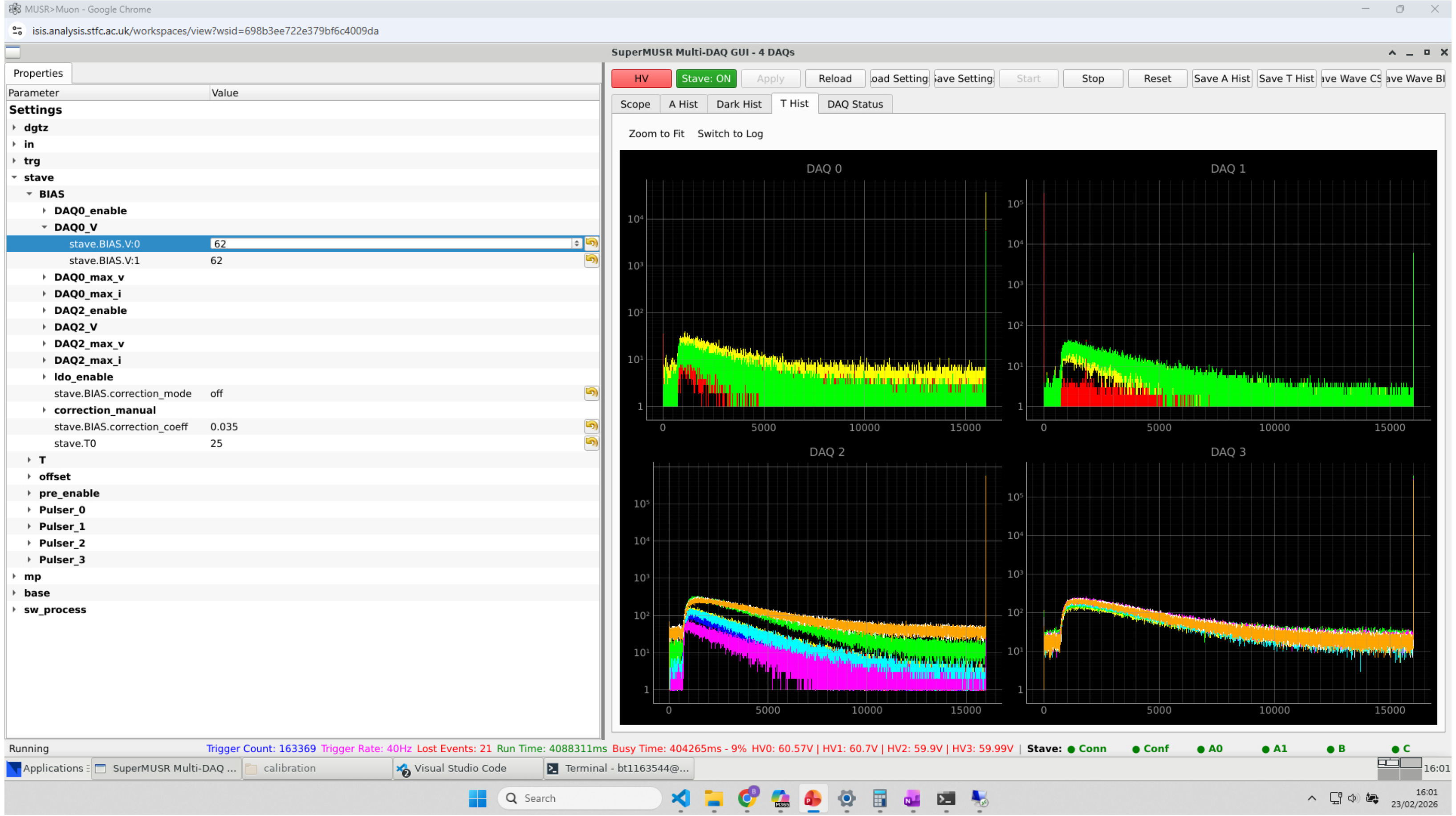
PCB: 20685
Total ID: 20689A006

PCB: 20688
Total ID: 20688A007

Magnetic Pins

Progress started with getting one full stave setup and running on beam (below, **STAVE 20718**), There is one broken cable, the LHS side one of CH3 on the MuSR Digitiser. Plot is below.

Faulty DAQ taken too the source room for investigation. Notes on how to update and how to use the calibration creation and HV selection GUI is noted. **When launching the GUI from the source room use “source_room_super_musr_codebase_260206”.** The other copy is in 4ch_gui under source room. This is a mistake and needs to be moved to the desktop and named something appropriate so there will not be confusion. This was tested on **STAVE 20720.**



STAVE PID:	A0 PID:	A1 PID:	B PID:	C PID:	Table Key:
20720	20688A005	20689A006	20689B002	20692C002	stave_conf_ 0002

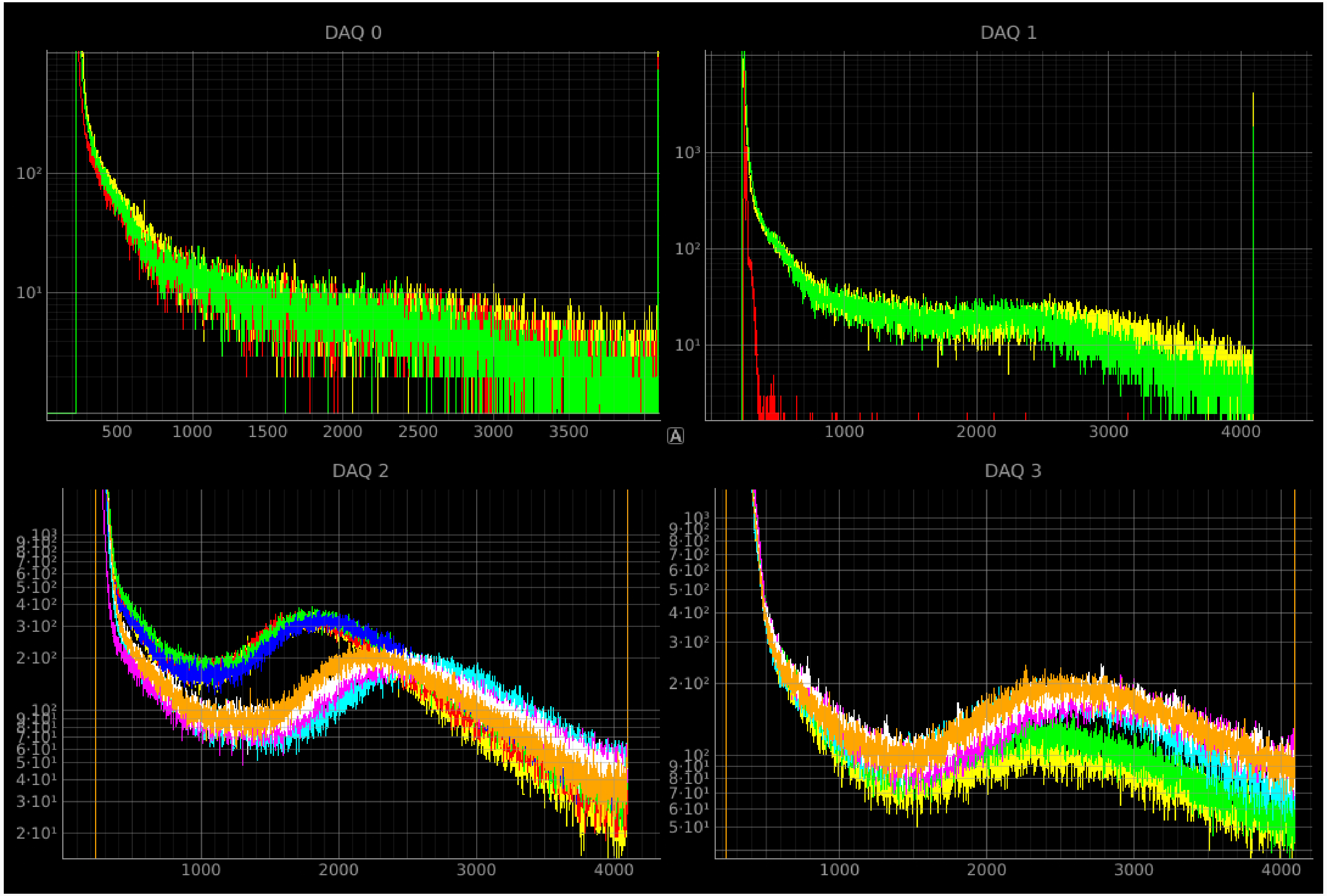
The issue with the broken DAQ was the infamous “stave_configuration=1”. Must be set to 2 for ALL SEVERS AND DIGITISERS. Below is screenshot from pulser of all channels working.

Stave PID: 20720

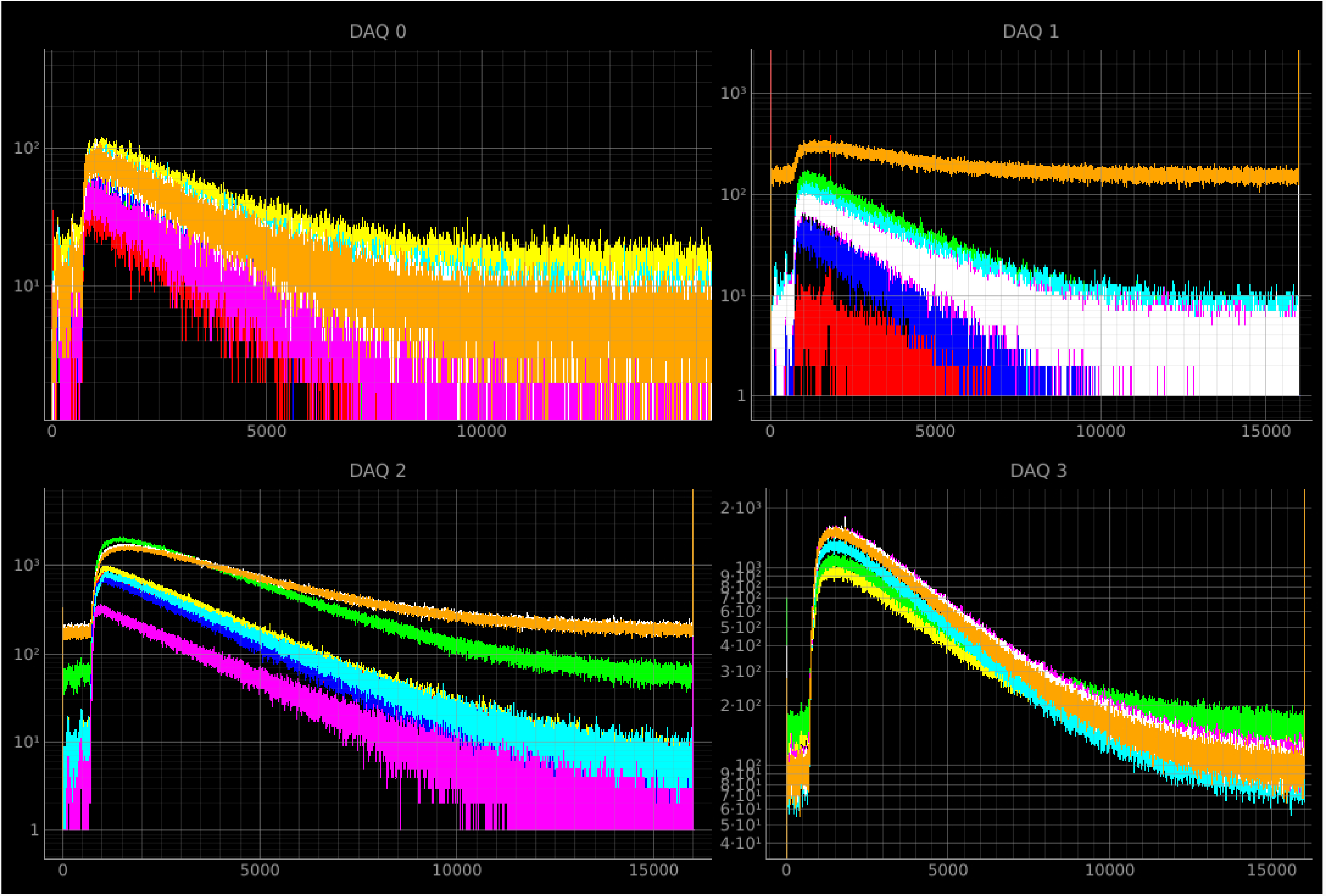
STAVE PID:	A0 PID:	A1 PID:	B PID:	C PID:	Table Key:
20720	20688A005	20689A006	20689B002	20692C002	stave_conf_ 0002

STAVE 20718

A0 PE	A1 PE	B PE	C PE
30	30	30	100



STAVE PID:	A0 PID:	A1 PID:	B PID:	C PID:	Table Key
20718	20686A003	20687A004	20690B003	20691C003	stv_conf_0001

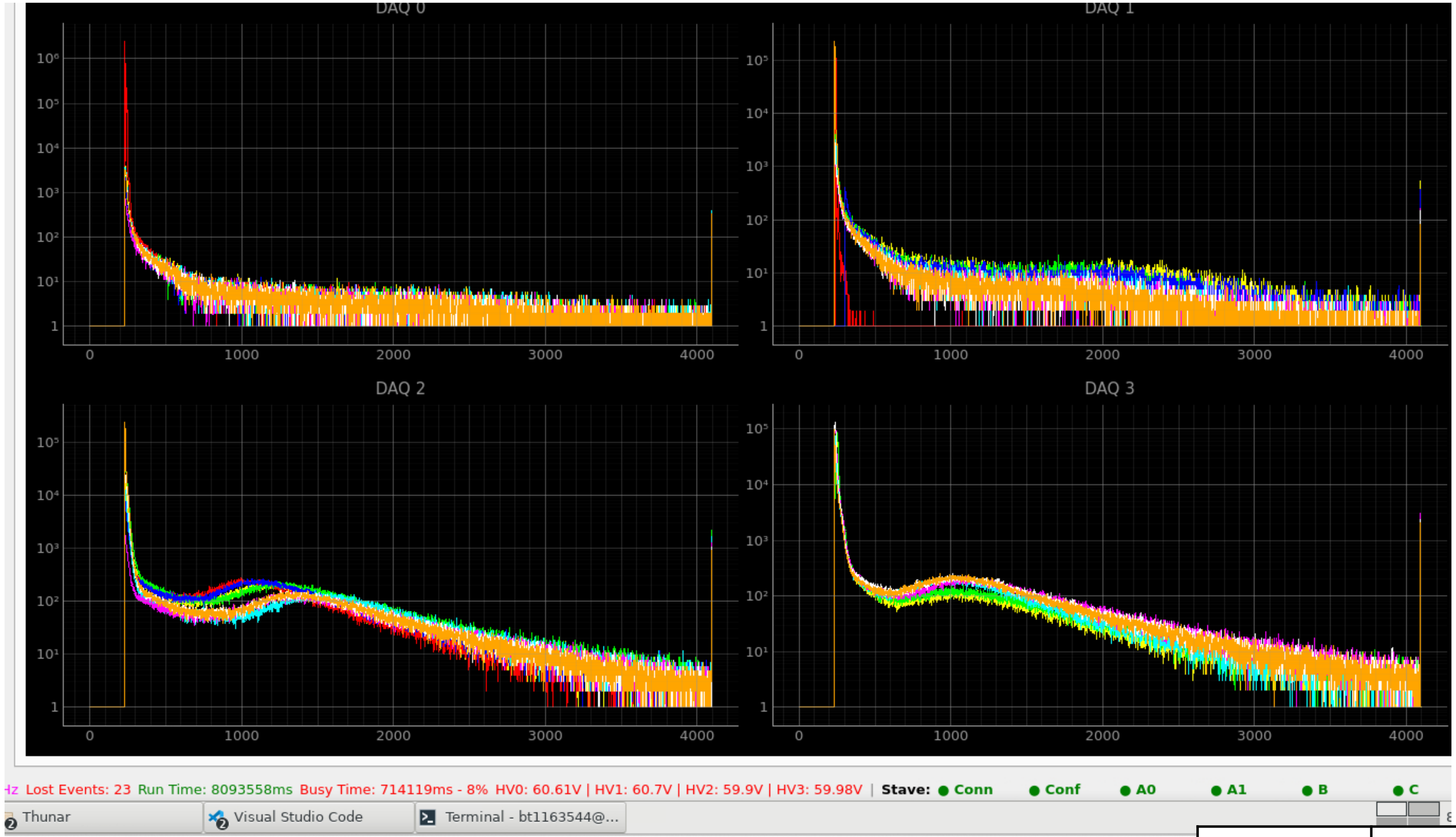


STAVE PID:	A0 PID:	A1 PID:	B PID:	C PID:	Table Key
20718	20686A003	20687A004	20690B003	20691C003	stv_conf_0001

A0 PE	A1 PE	B PE	C PE
30	30	30	100

PE 25 25 25 50

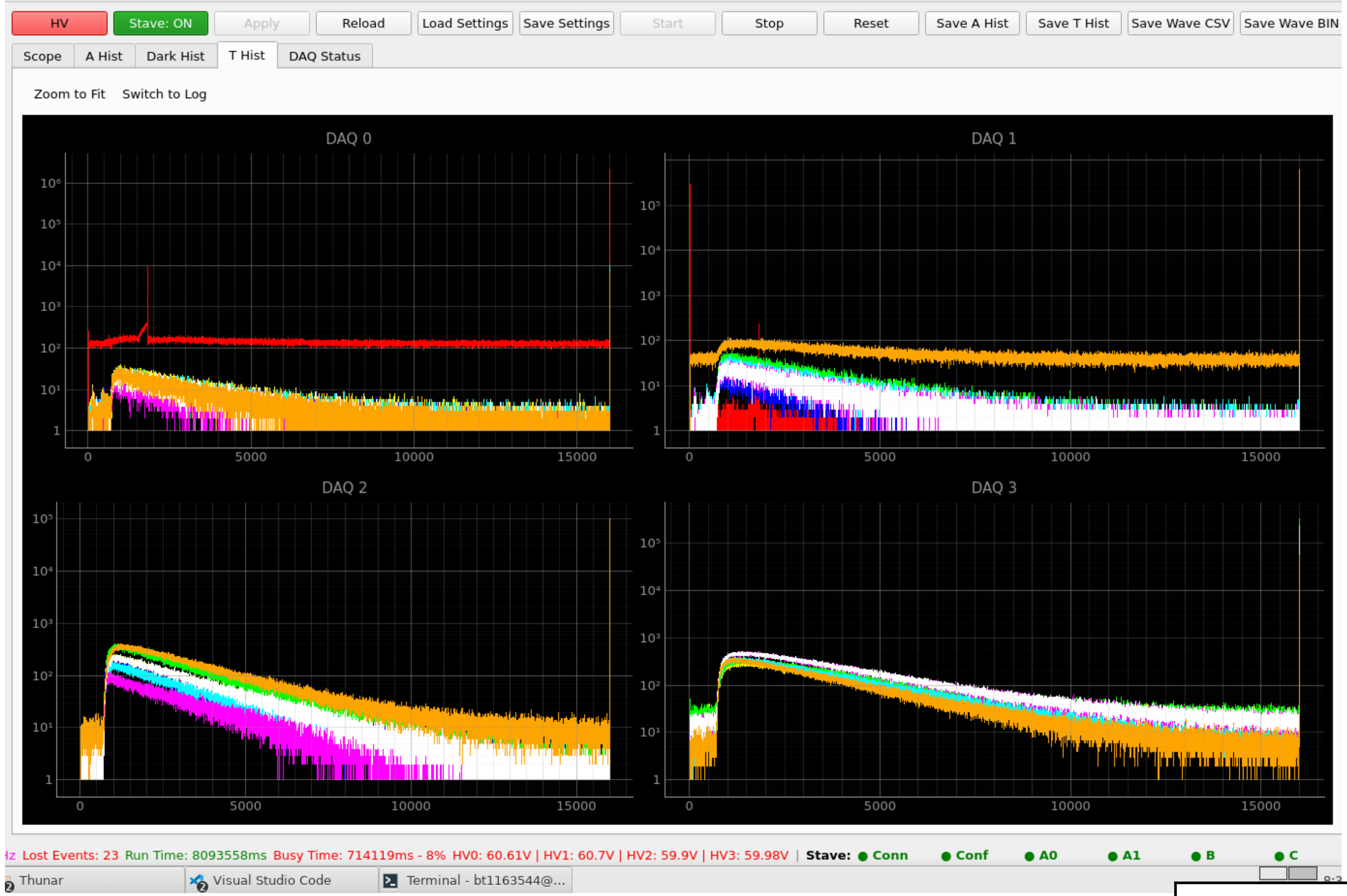
geometry set to have modules B and C exposed to the source



STAVE PID:	A0 PID:	A1 PID:	B PID:	C PID:	Table Key
20718	20686A003	20687A004	20690B003	20691C003	stv_conf_0001

A0 PE	A1 PE	B PE	C PE
25	25	25	50

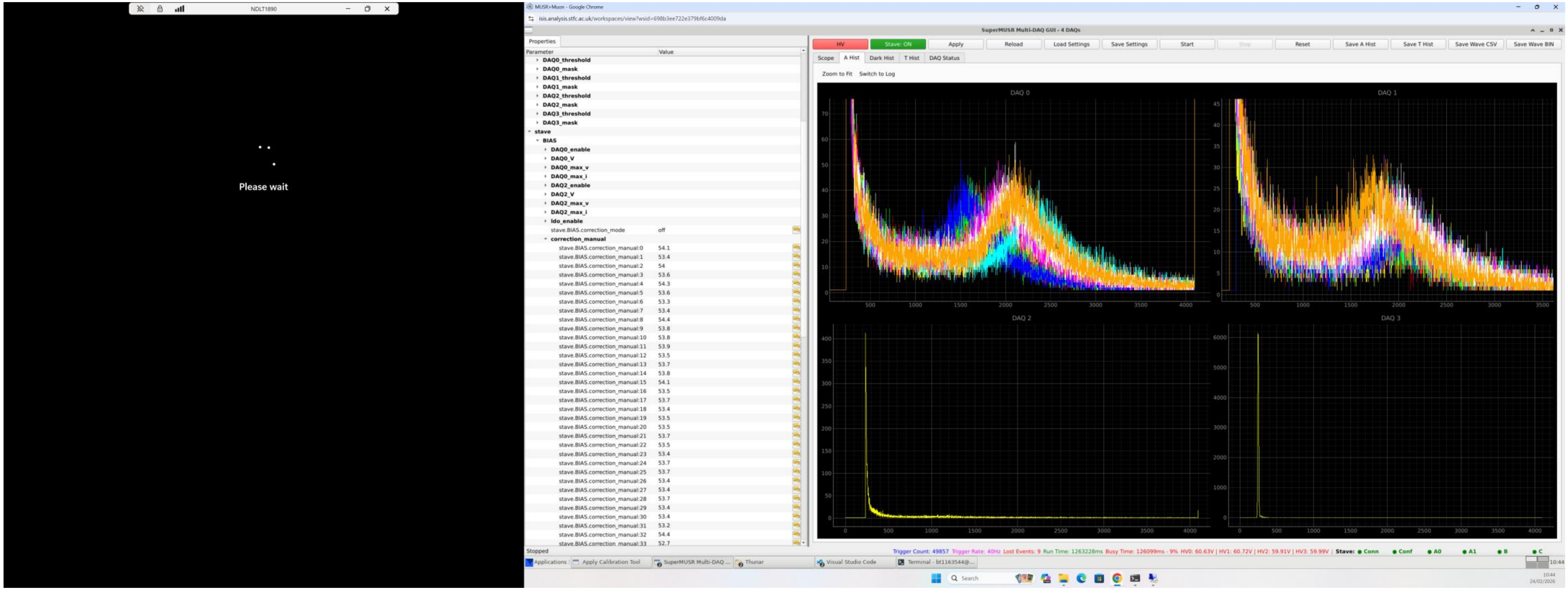
geometry set to have modules B and C exposed to the source



STAVE PID:	A0 PID:	A1 PID:	B PID:	C PID:	Table Key
20718	20686A003	20687A004	20690B003	20691C003	stv_conf_0001

A0 PE	A1 PE	B PE	C PE
25	25	25	50

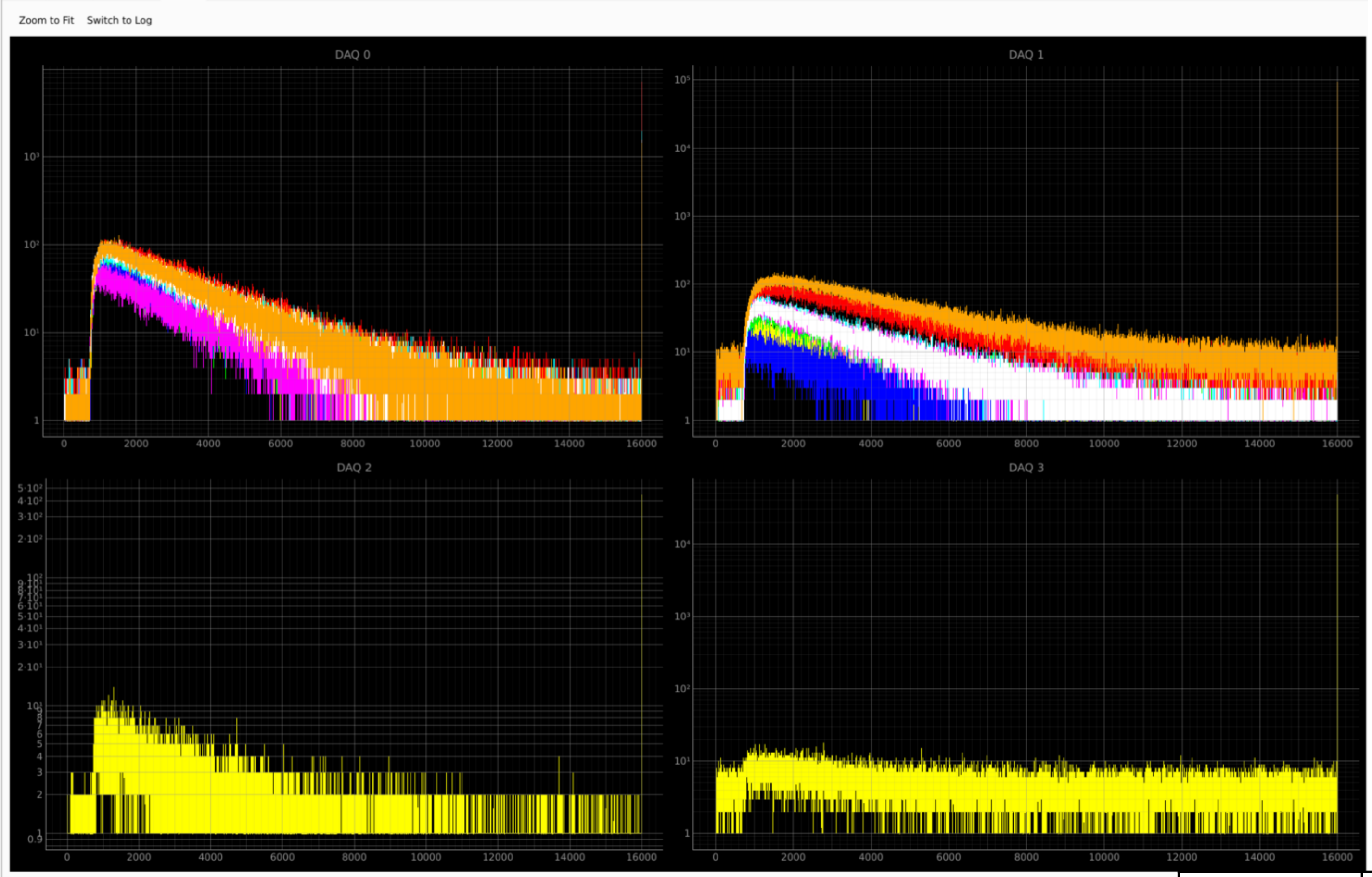
geometry set to have modules A0 and A1 exposed to the source



STAVE PID:	A0 PID:	A1 PID:	B PID:	C PID:	Table Key
20718	20686A003	20687A004	20690B003	20691C003	stv_conf_0001

A0 PE	A1 PE	B PE	C PE
20	20	20	50

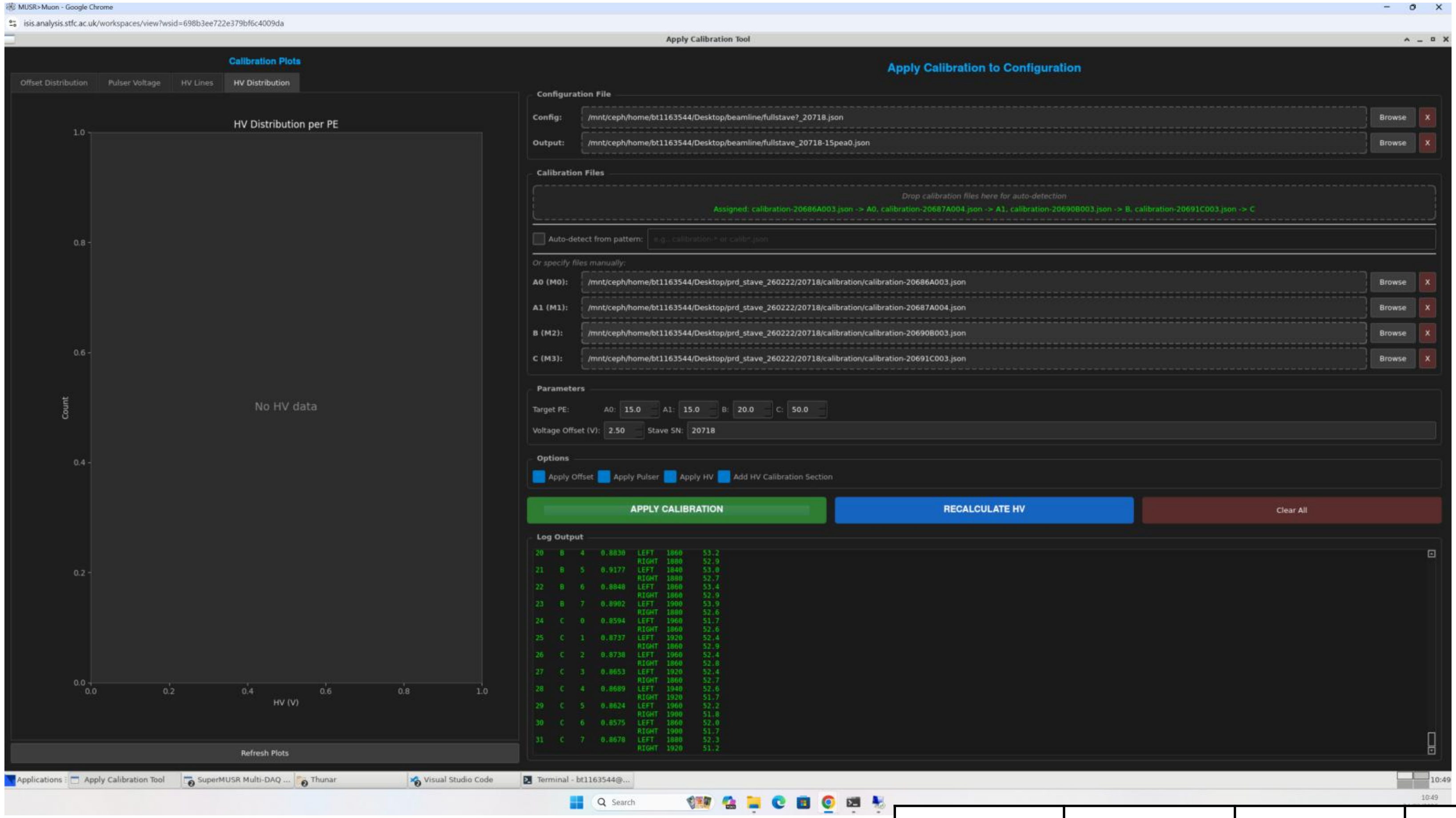
geometry set to have modules A0 and A1 exposed to the source



A0 PE	A1 PE	B PE	C PE
20	20	20	50

STAVE PID:	A0 PID:	A1 PID:	B PID:	C PID:	Table Key
20718	20686A003	20687A004	20690B003	20691C003	stv_conf_0001

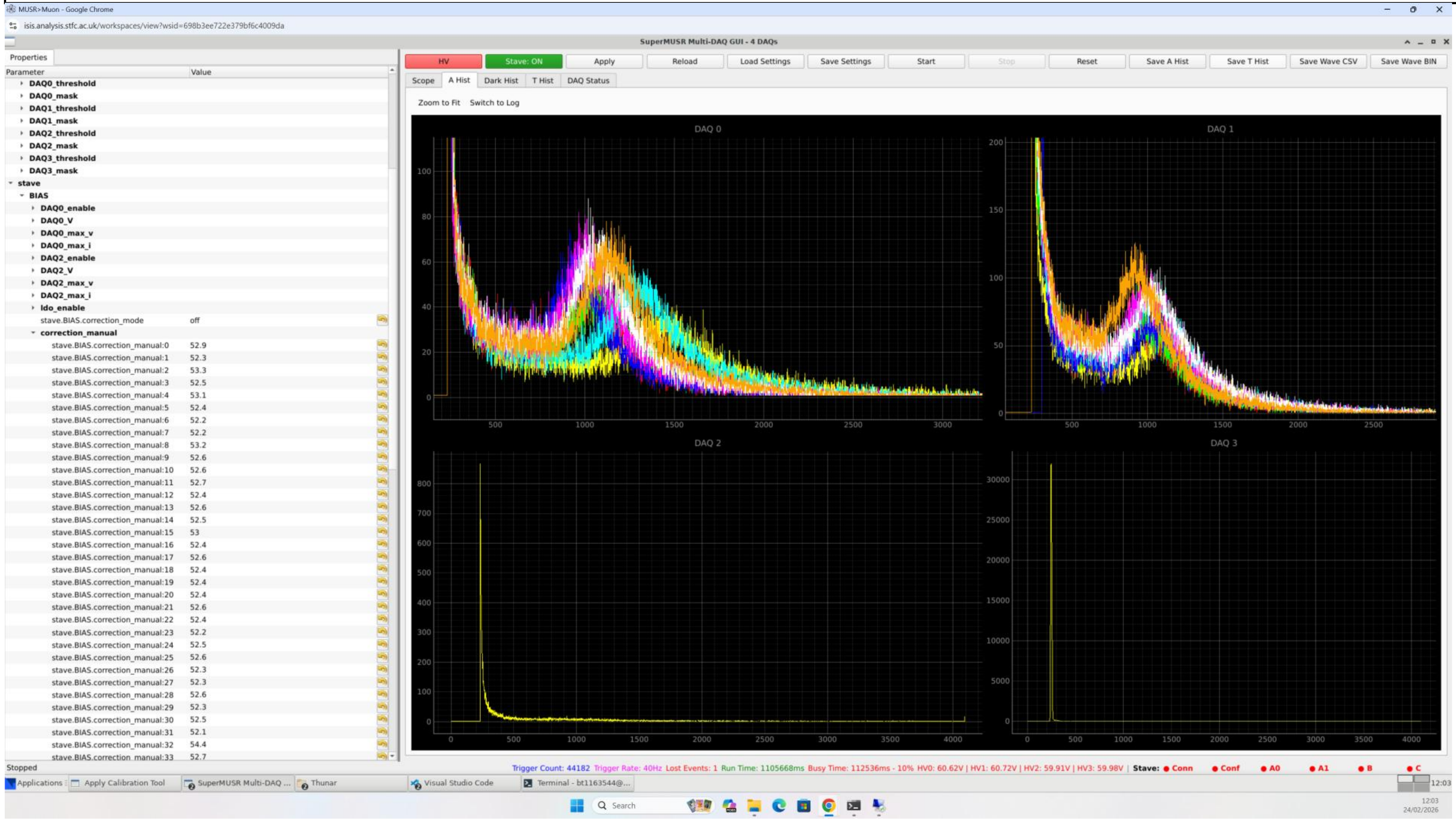
This is the calibration file set running for 24/02/26 10:45 onwards with geometry set to have modules A0 and A1 exposed to the source



STAVE PID:	A0 PID:	A1 PID:	B PID:	C PID:	Table Key
20718	20686A003	20687A004	20690B003	20691C003	stv_conf_0001

A0 PE	A1 PE	B PE	C PE
15	15	20	50

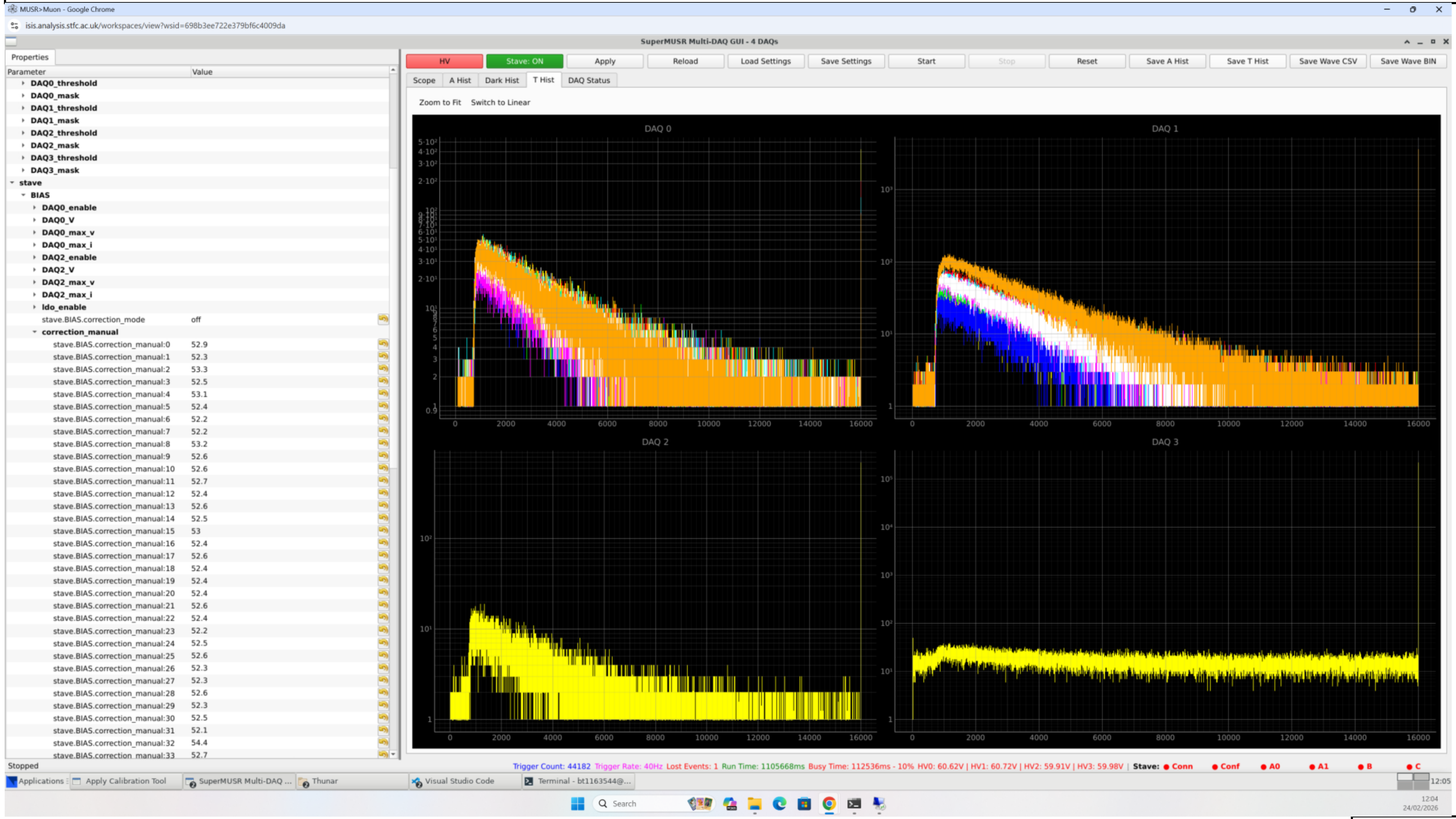
File path on IdaaAS: " /home/bt1163544\Desktop\beamline\run0004 – 15 20 15 50". PE is a0:15, a1:15, b:20, c:50. Geometry is focussed on modules a0 & a1. Laying aprox 45 degree on table, 3mm brass degrader, shifted so that positron direction is aligned with a0 & a1. Bin waves etc saved.



STAVE PID:	A0 PID:	A1 PID:	B PID:	C PID:	Table Key
20718	20686A003	20687A004	20690B003	20691C003	stv_conf_0001

A0 PE	A1 PE	B PE	C PE
15	15	20	50

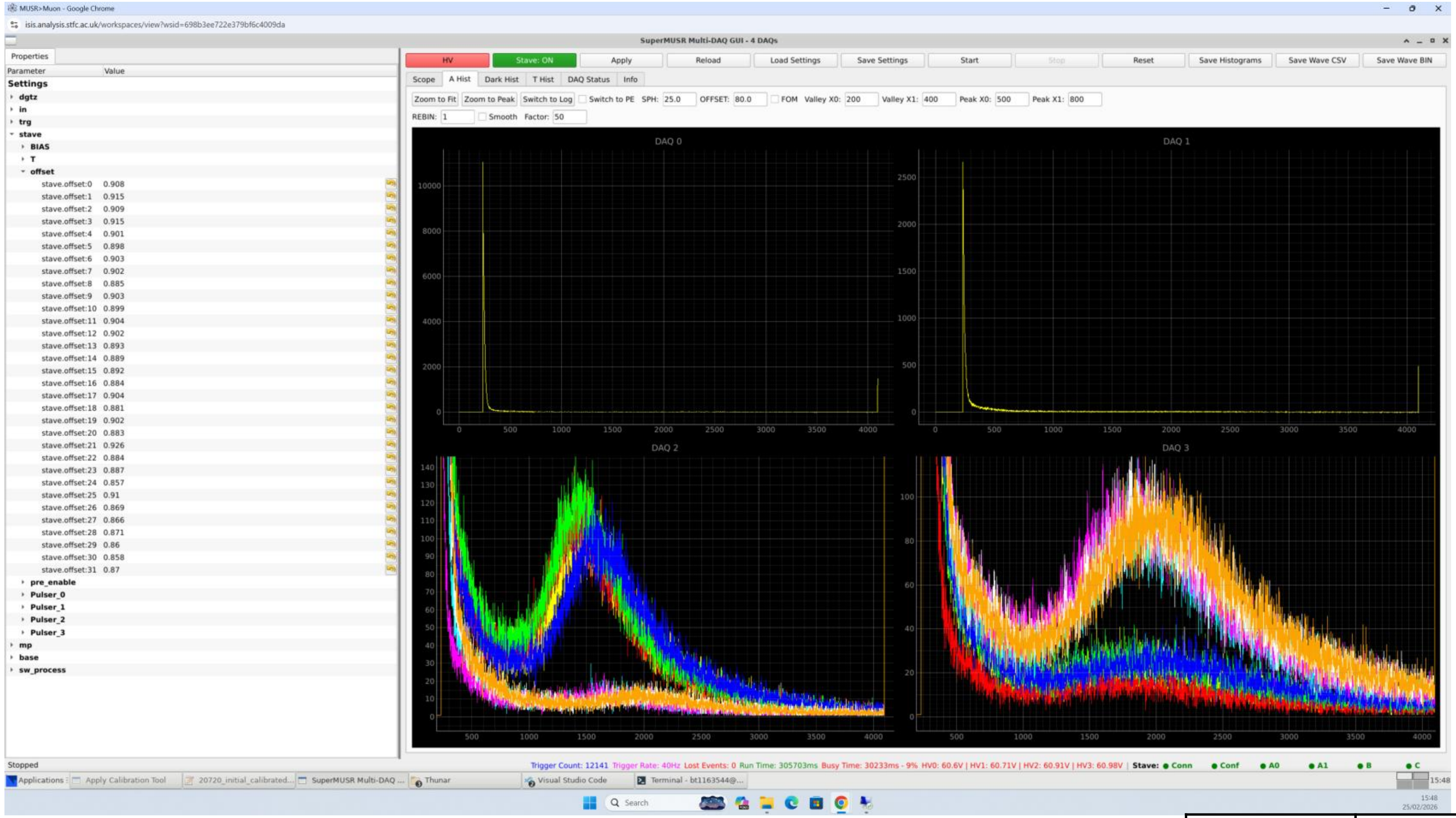
File path on IdaaAS: " /home/bt1163544\Desktop\beamline\run0004 – 15 20 15 50". PE is a0:15, a1:15, b:20, c:50. Geometry is focussed on modules a0 & a1. Laying aprox 45 degree on table, 3mm brass degrader, shifted so that positron direction is aligned with a0 & a1. Bin waves etc saved.



STAVE PID:	A0 PID:	A1 PID:	B PID:	C PID:	Table Key
20718	20686A003	20687A004	20690B003	20691C003	stv_conf_0001

A0 PE	A1 PE	B PE	C PE
15	15	20	50

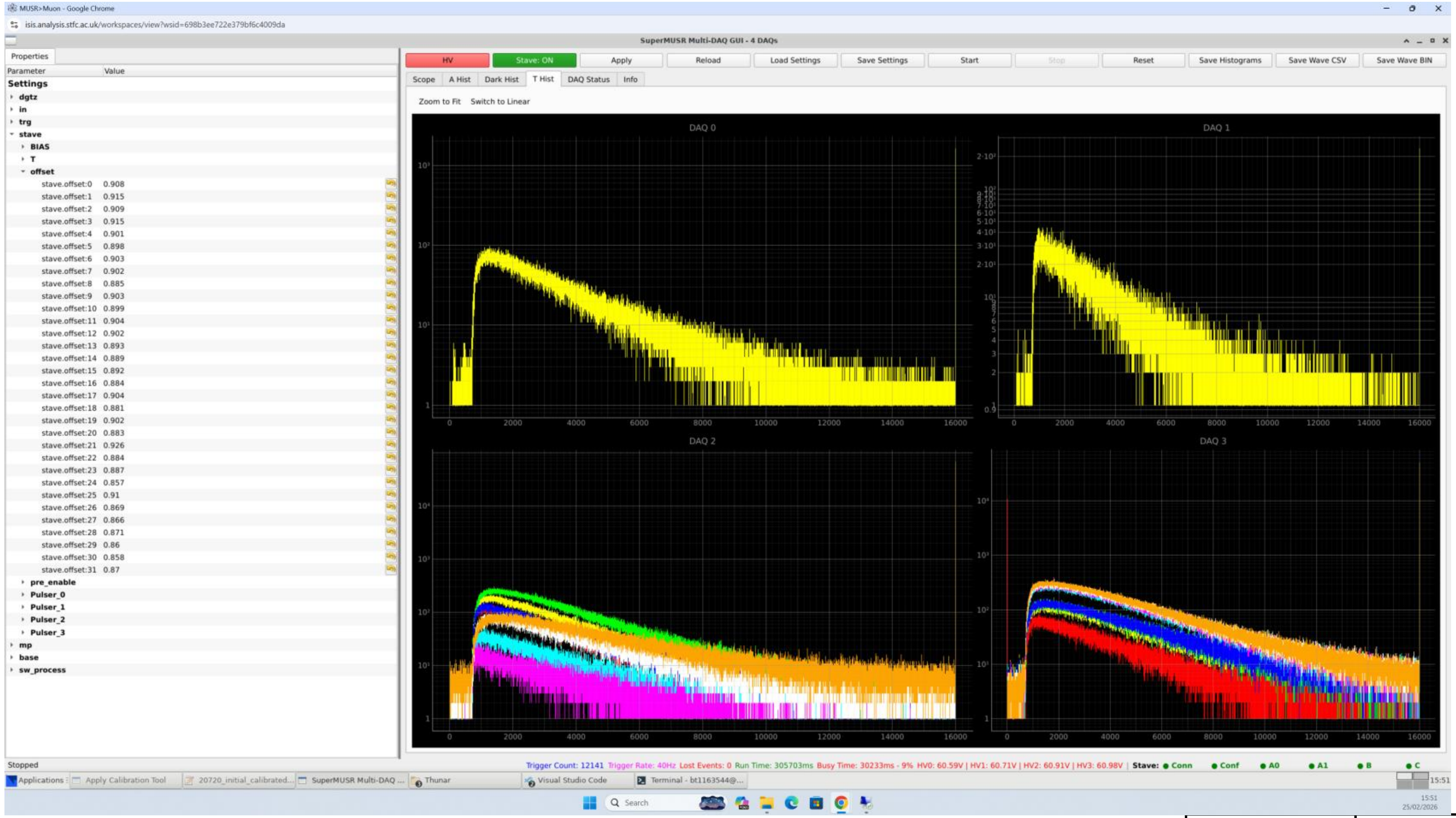
File path on IdaaAS: " /home/bt1163544\Desktop\beamline\run0008 – 25 25 35 75". PE is a0:25, a1:25 b:35, c:75. Geometry is focussed on modules c. Laying aprox 30 degrees. Binary waves taken. Note this not optimal positioning, no Perspex shim etc. data. Note that offset for ch1 on Module C had an incorrect offset, so this is why the PHS has lower stats.



STAVE PID:	A0 PID:	A1 PID:	B PID:	C PID:	Table Key:
20720	20688A005	20689A006	20689B002	20692C002	stave_conf_0002

A0 PE	A1 PE	B PE	C PE
25	25	35	75

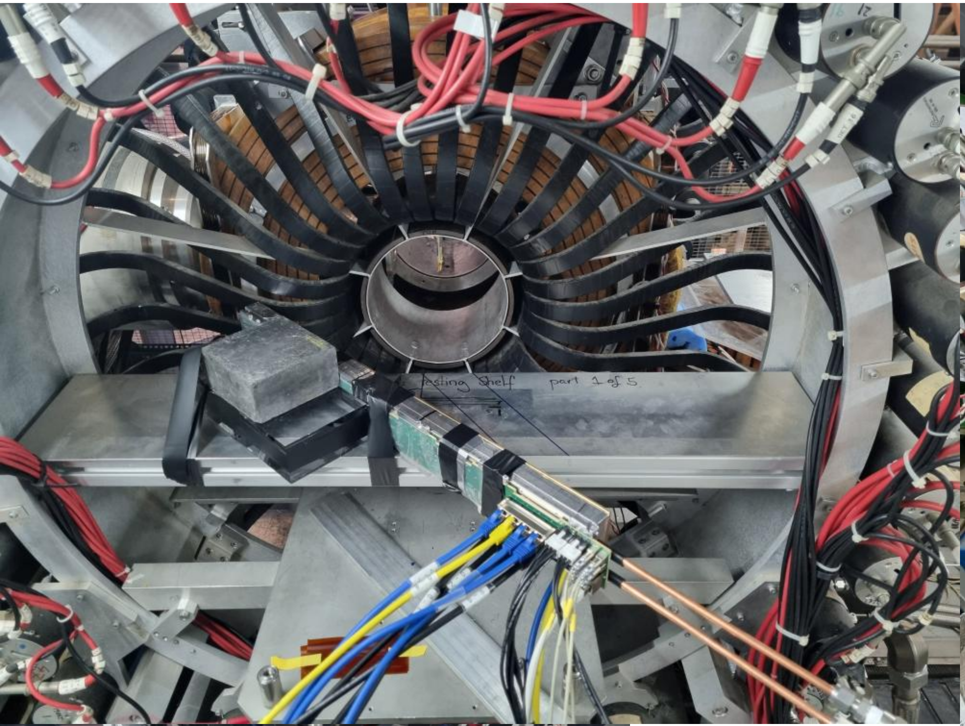
File path on IdaaAS: " /home/bt1163544\Desktop\beamline\run0008 – 25 25 35 75". PE is a0:25, a1:25 b:35, c:75. Geometry is focussed on modules c. Laying aprox 30 degrees. Binary waves taken. Note this not optimal positioning, no Perspex shim etc. data. Note that offset for ch1 on Module C had an incorrect offset, so this is why the PHS has lower stats.

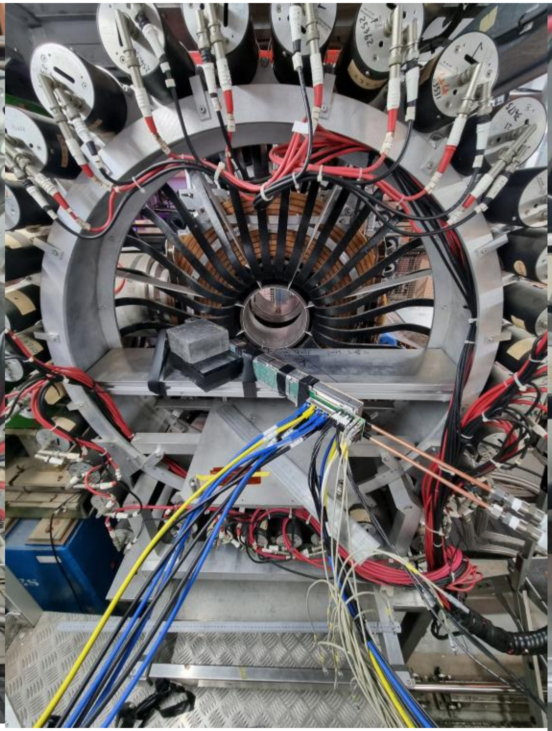


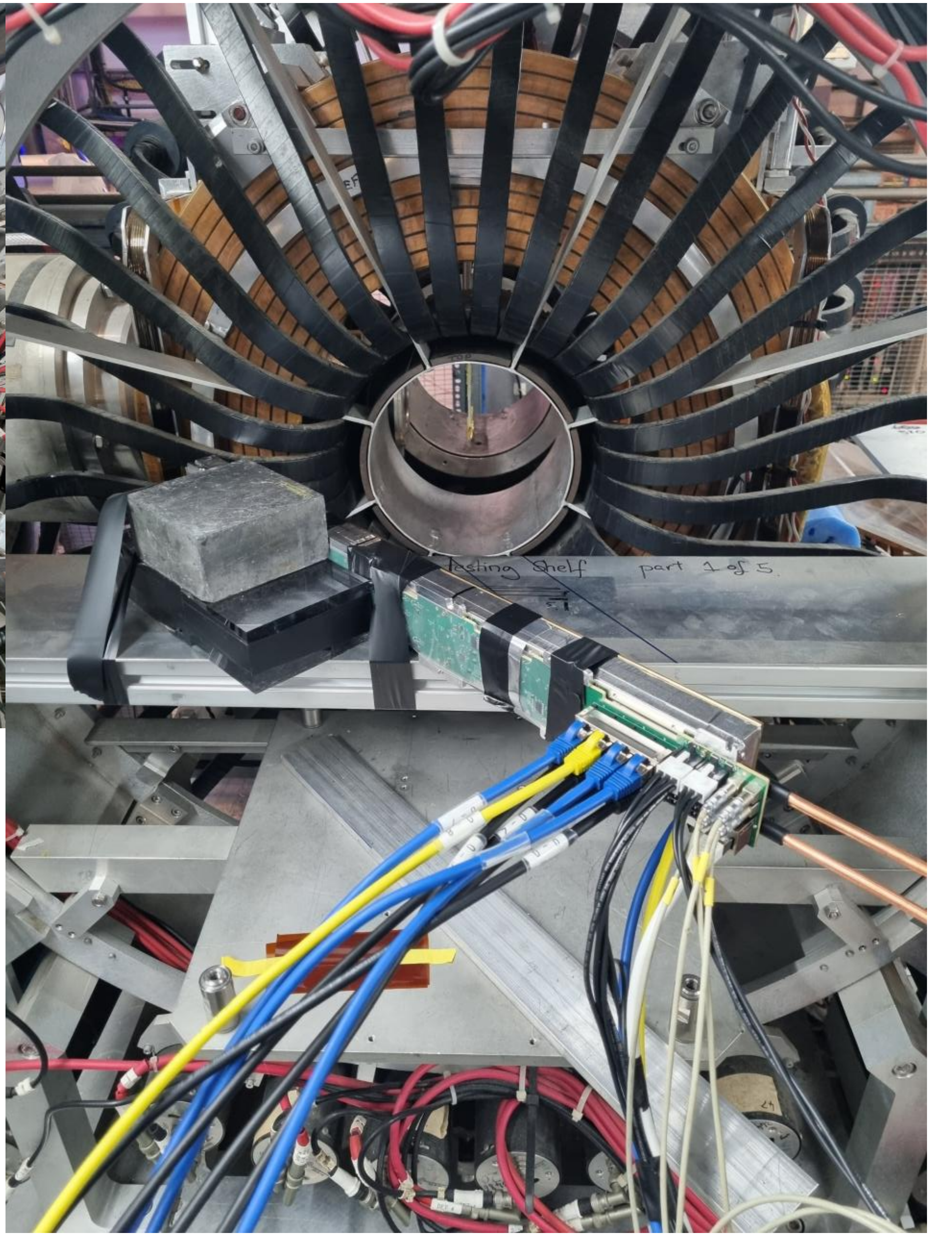
STAVE PID:	A0 PID:	A1 PID:	B PID:	C PID:	Table Key:
20720	20688A005	20689A006	20689B002	20692C002	stave_conf_0002

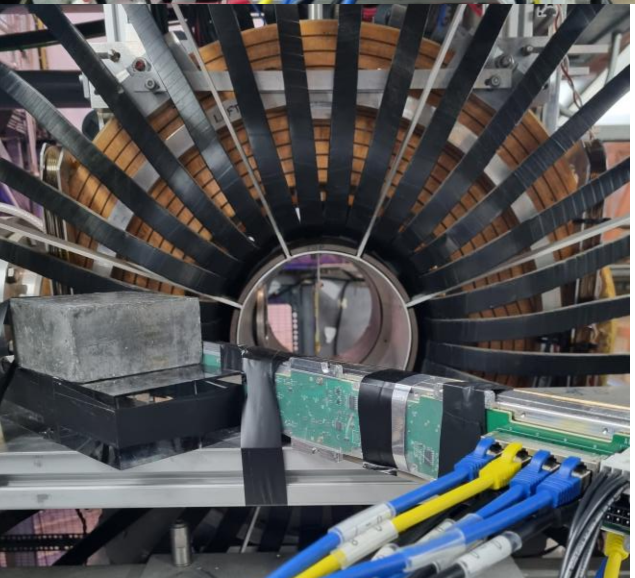
A0 PE	A1 PE	B PE	C PE
25	25	35	75

Full stave Module B geometry example

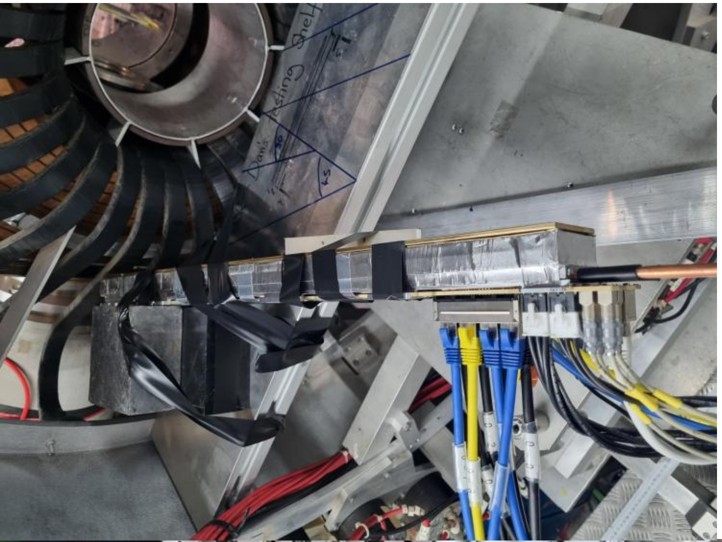


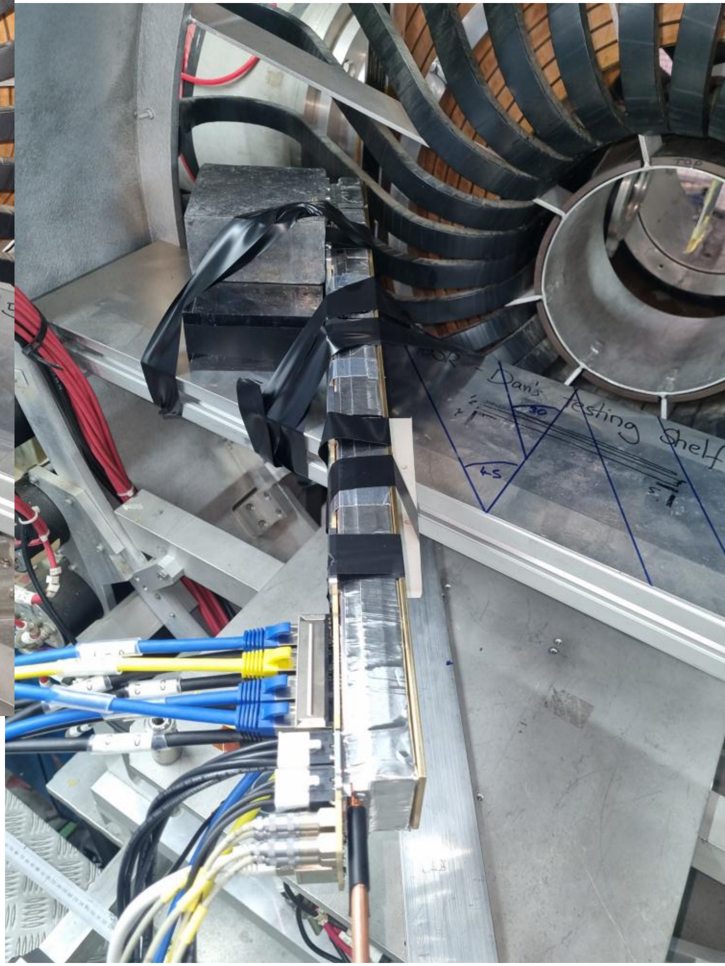


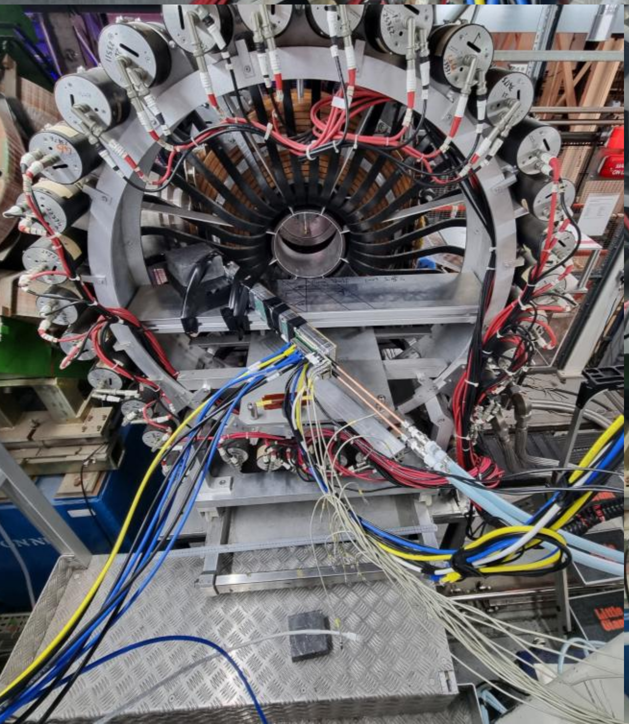


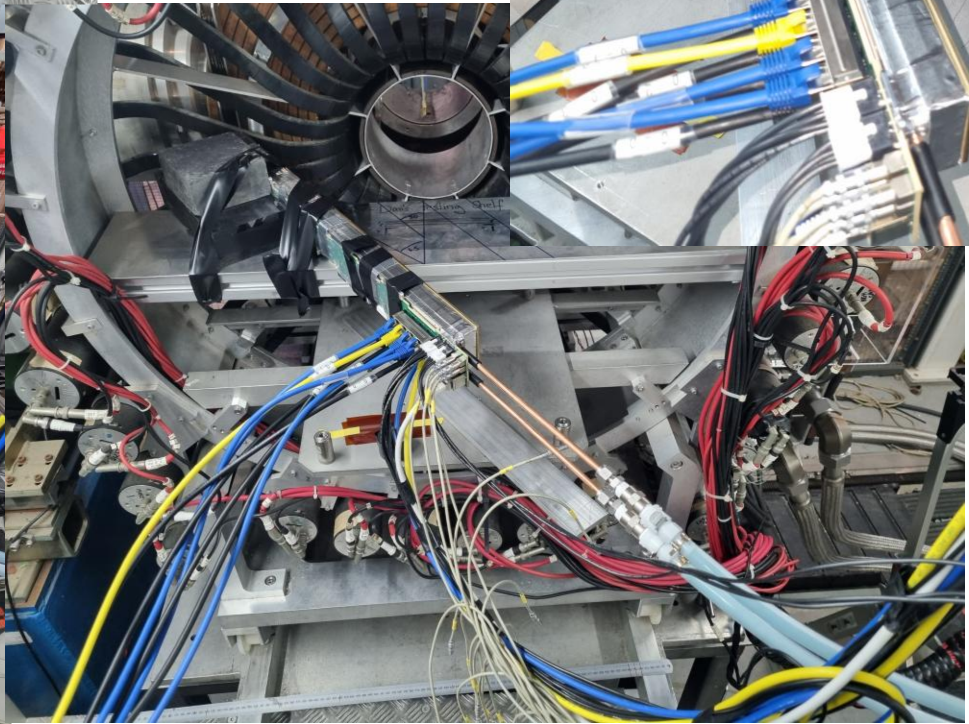


Full stave Module C geometry example







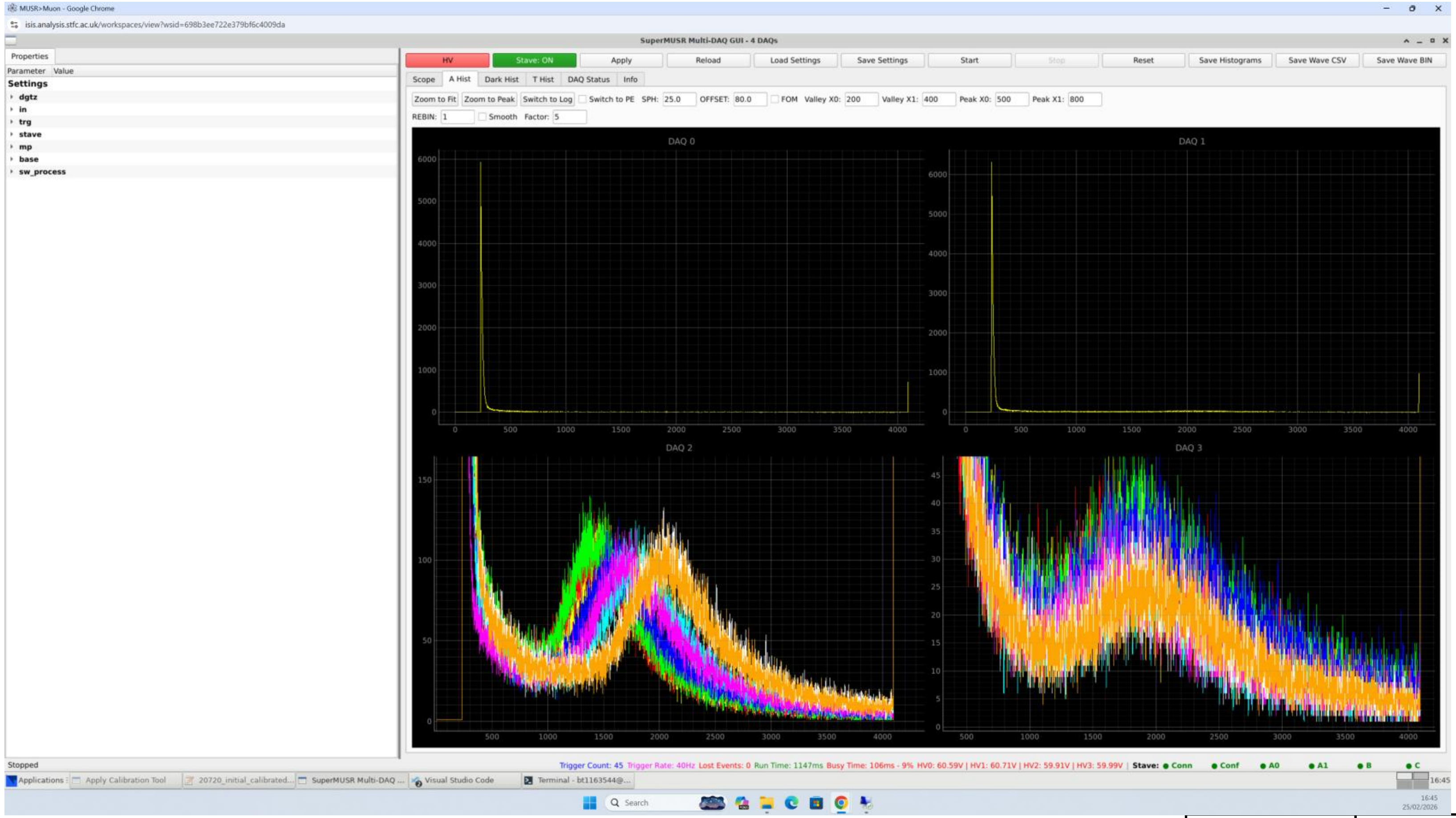


Conclusions – Data rate Kafka Compression

- No compression.
  - We saw no dropped frames on DAQ, or from Broker at 16us and 32us acquisition, with 1 and 8 DAQS running. It works without failure.
  - One broker CPU@20%, Netowrk TX@160Mbit/s (40Hz, 32us).
- zstd (On one DAQ)
  - Compression ratio ~x2, seen as TX going from [Mbit/s]
- Lz4 (On one DAQ)
  - Compression ratio ~x, seen as TX going from [Mbit/s]

- Compression adds CPU overhead at both ends. The broker is overwhelmed and returns error

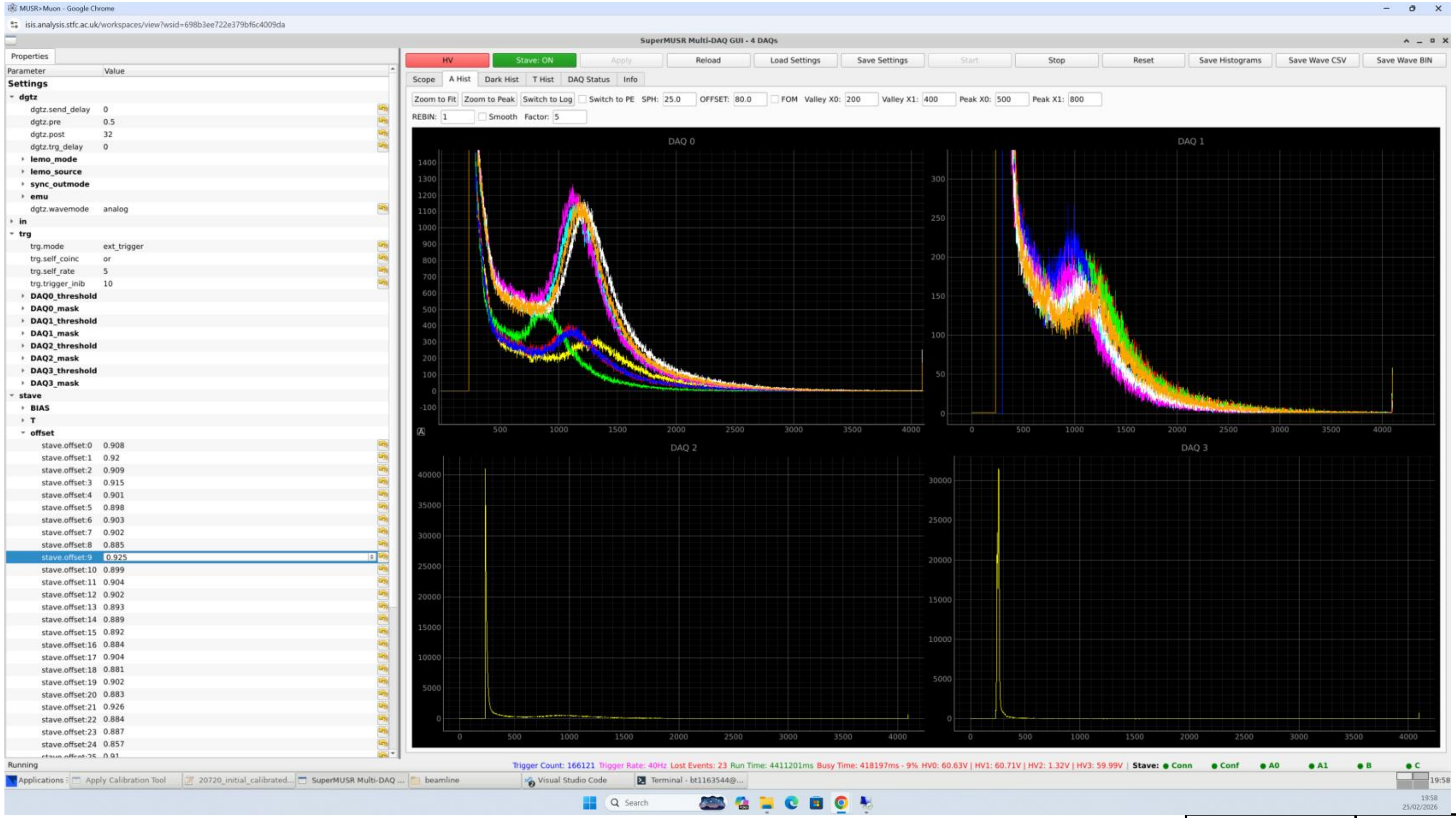
File path on IdaaAS: “ /home/bt1163544\Desktop\beamline\run0010 – 25 25 35 75”. PE is a0:25, a1:25 b:35, c:75. Geometry is focussed on modules B . Laying approx. 48 degrees. Binary waves taken. Bad picture as the GUI crashed.



A0 PE	A1 PE	B PE	C PE
25	25	35	75

STAVE PID:	A0 PID:	A1 PID:	B PID:	C PID:	Table Key:
20720	20688A005	20689A006	20689B002	20692C002	stave_conf_0002

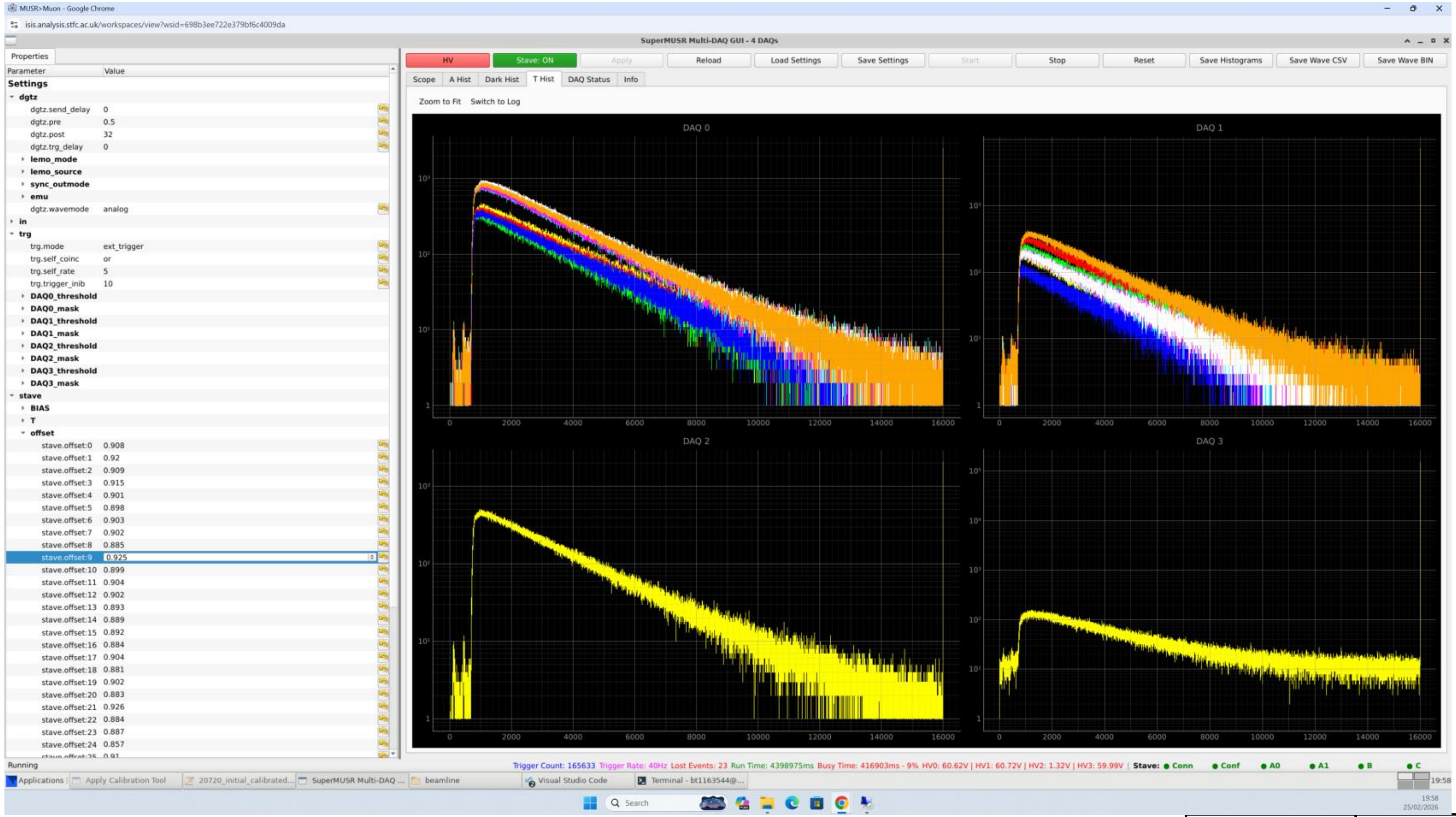
File path on IdaaAS: “ /home/bt1163544\Desktop\beamline\run0011 – 15 15 25 50”. PE is a0:15, a1:215, b:25, c:50. Geometry is focussed on modules A0.



STAVE PID:	A0 PID:	A1 PID:	B PID:	C PID:	Table Key:
20720	20688A005	20689A006	20689B002	20692C002	stave_conf_0002

A0 PE	A1 PE	B PE	C PE
15	15	25	50

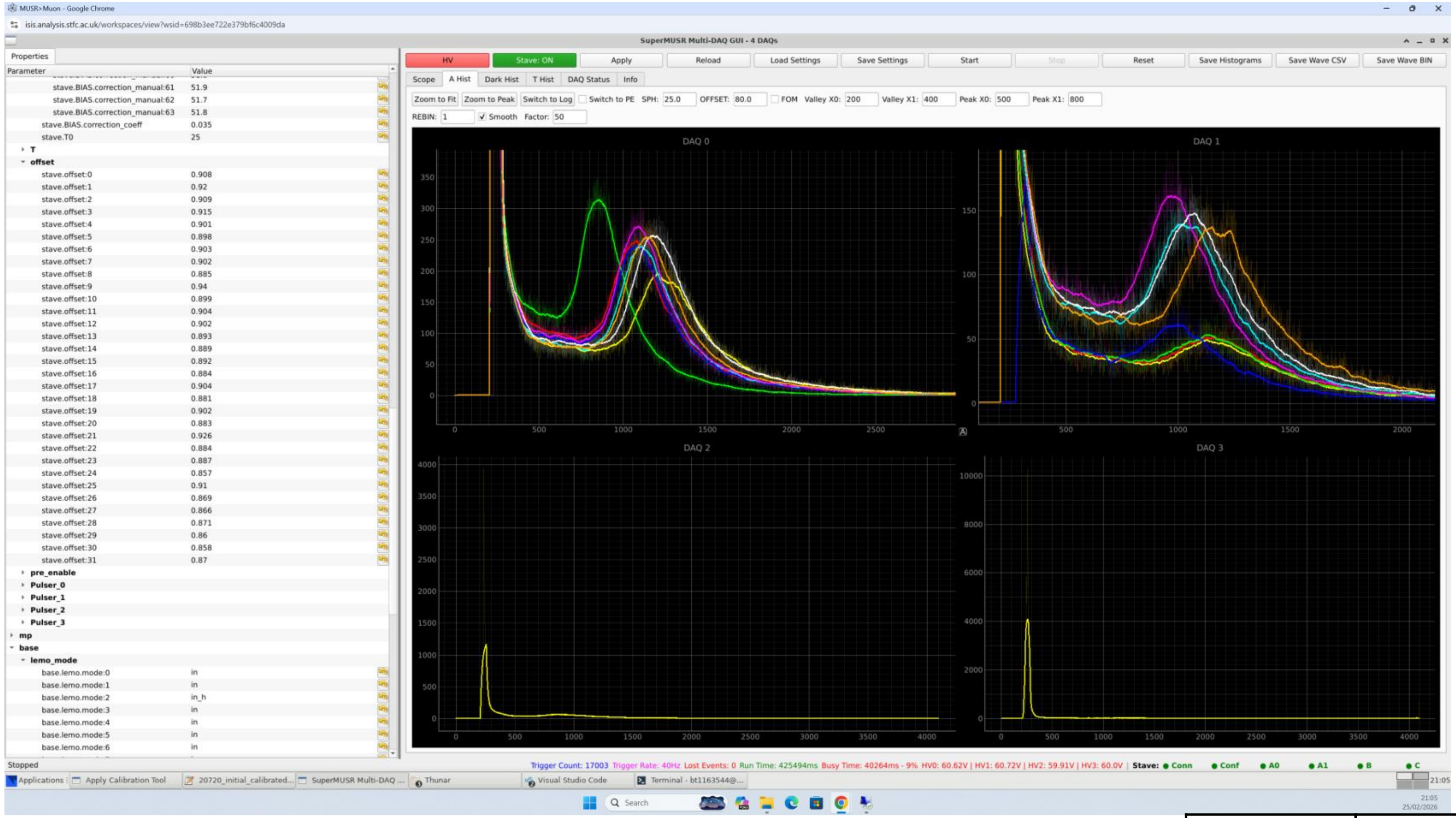
File path on IdaaAS: “ /home/bt1163544\Desktop\beamline\run0011 – 15 15 25 50”. PE is a0:15, a1:215, b:25, c:50. Geometry is focussed on modules A0.



STAVE PID:	A0 PID:	A1 PID:	B PID:	C PID:	Table Key:
20720	20688A005	20689A006	20689B002	20692C002	stave_conf_0002

A0 PE	A1 PE	B PE	C PE
15	15	25	50

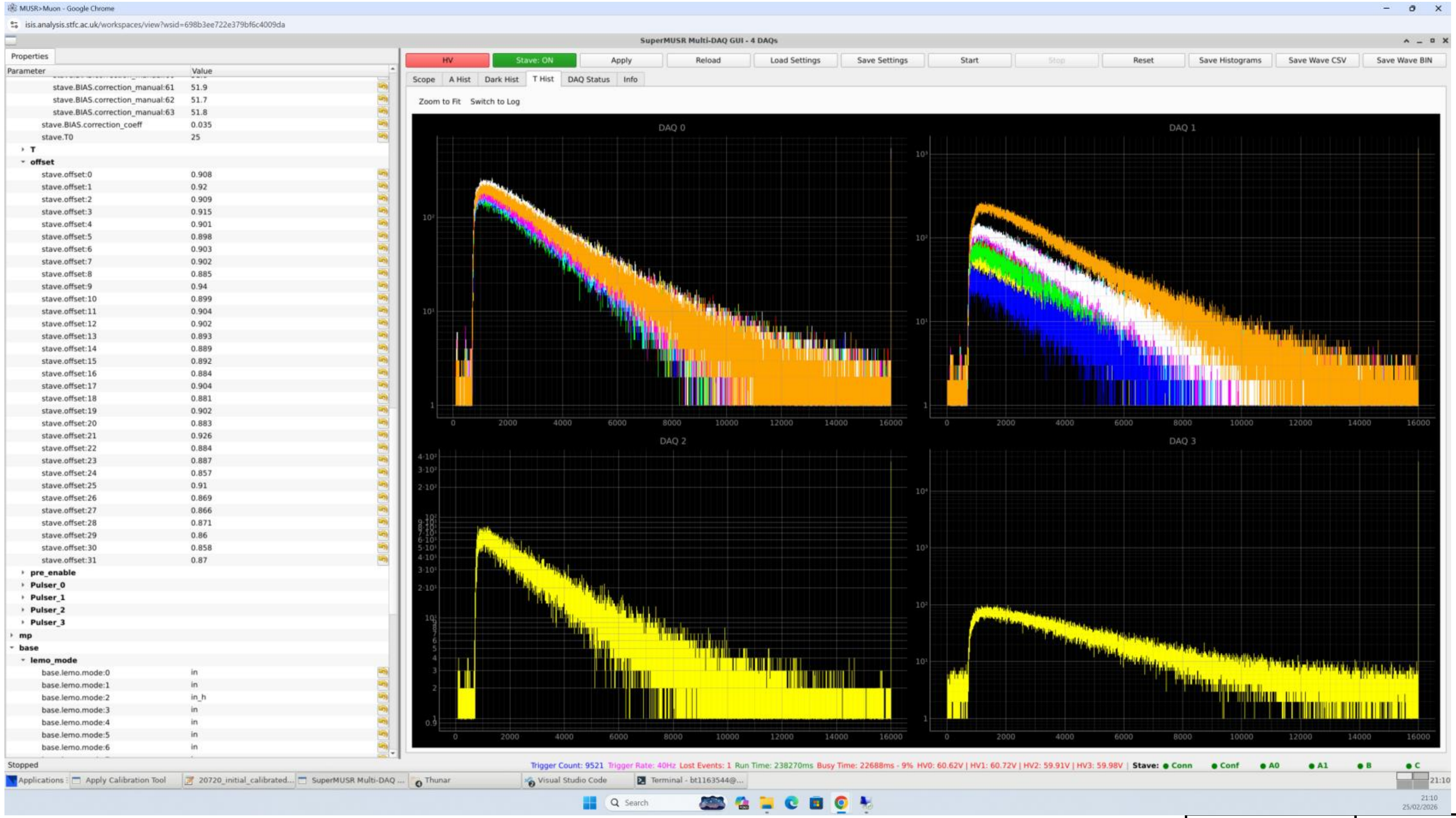
Odd channel behaviour on ch2, seems to have about half of the PHS peak. File path on IdaaAS: “ /home/bt1163544\Desktop\beamline\run0011 – 15 15 25 50”. PE is a0:15, a1:215, b:25, c:50. Geometry is focussed on modules A0. Odd channel behaviour on ch2, seems to have about half of the PHS peak. Tile is to be swapped.



STAVE PID:	A0 PID:	A1 PID:	B PID:	C PID:	Table Key:
20720	20688A005	20689A006	20689B002	20692C002	stave_conf_0002

A0 PE	A1 PE	B PE	C PE
15	15	25	50

Odd channel behaviour on ch2, seems to have about half of the PHS peak. File path on IdaaAS: “ /home/bt1163544\Desktop\beamline\run0011 – 15 15 25 50”. PE is a0:15, a1:215, b:25, c:50. Geometry is focussed on modules A0. Odd channel behaviour on ch2, seems to have about half of the PHS peak. Tile is to be swapped.



STAVE PID:	A0 PID:	A1 PID:	B PID:	C PID:	Table Key:
20720	20688A005	20689A006	20689B002	20692C002	stave_conf_0002

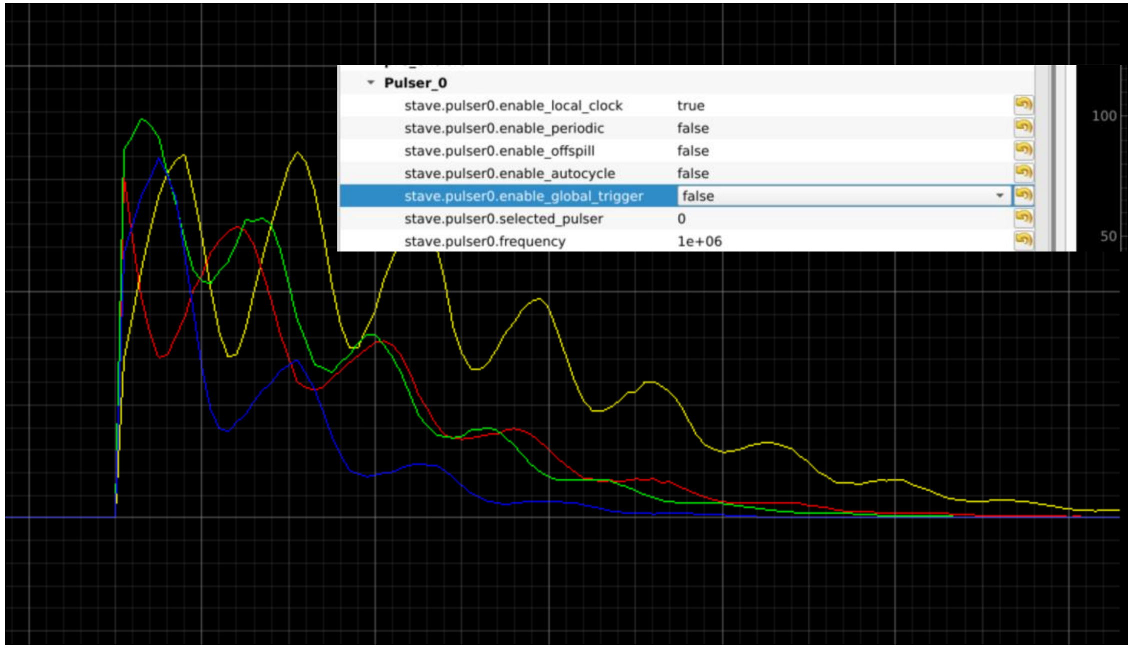
A0 PE	A1 PE	B PE	C PE
15	15	25	50

We investigated ch 2 on mod A0. It has a PHS with the peak downshifted with respect to the others in the same ring. To understand this ben swapped two tiles over. stv_conf_003

There was some improvement (peak shifted to higher LSB) but clearly still lower light collection.

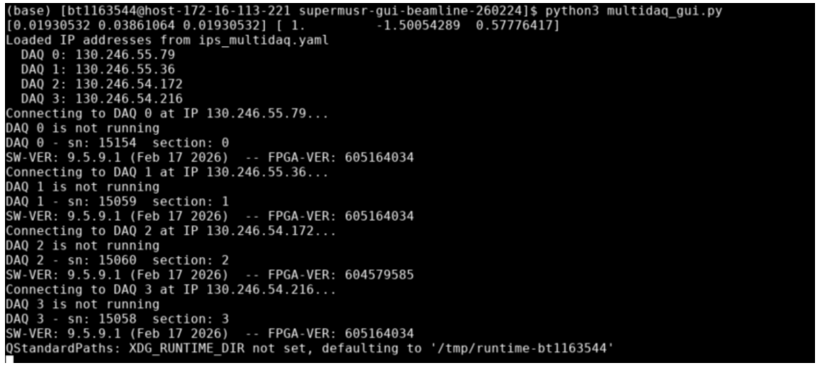
We checked both sipms were on and gain matched using the pulser mps. (out of spill).

Confirmed both SiPMs are on. Because the other channels were unaffected by the swap, it appears to be a real misalignment and light loss.



GUI freeze.

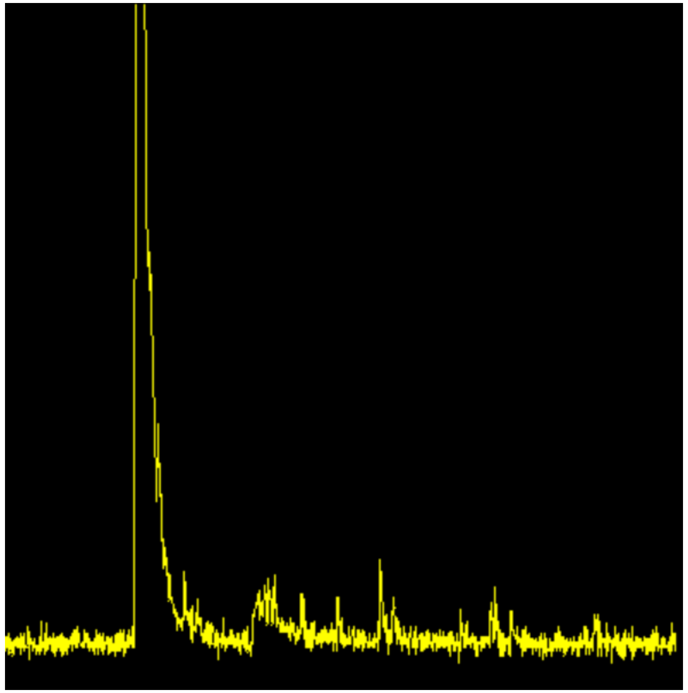
I save settings.



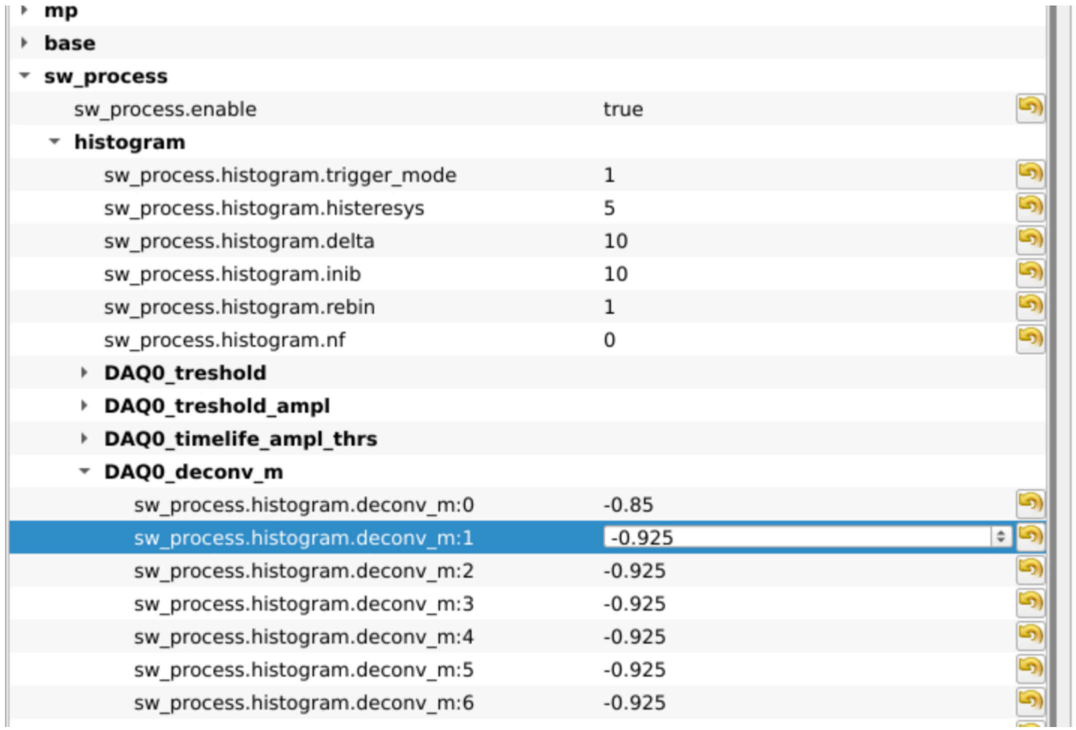
```
(base) [bt1163544@host-172-16-113-221 supermusr-gui-beamline-260224]$ python3 multidaq_gui.py
[0.01930532 0.03861064 0.01930532] [ 1.         -1.50054289  0.57776417]
Loaded IP addresses from ips_multidaq.yaml
  DAQ 0: 130.246.55.79
  DAQ 1: 130.246.55.36
  DAQ 2: 130.246.54.172
  DAQ 3: 130.246.54.216
Connecting to DAQ 0 at IP 130.246.55.79...
DAQ 0 is not running
DAQ 0 - sn: 15154  section: 0
SW-VER: 9.5.9.1 (Feb 17 2026)  -- FPGA-VER: 605164034
Connecting to DAQ 1 at IP 130.246.55.36...
DAQ 1 is not running
DAQ 1 - sn: 15059  section: 1
SW-VER: 9.5.9.1 (Feb 17 2026)  -- FPGA-VER: 605164034
Connecting to DAQ 2 at IP 130.246.54.172...
DAQ 2 is not running
DAQ 2 - sn: 15060  section: 2
SW-VER: 9.5.9.1 (Feb 17 2026)  -- FPGA-VER: 604579585
Connecting to DAQ 3 at IP 130.246.54.216...
DAQ 3 is not running
DAQ 3 - sn: 15058  section: 3
SW-VER: 9.5.9.1 (Feb 17 2026)  -- FPGA-VER: 605164034
QStandardPaths: XDG_RUNTIME_DIR not set, defaulting to '/tmp/runtime-bt1163544'
```

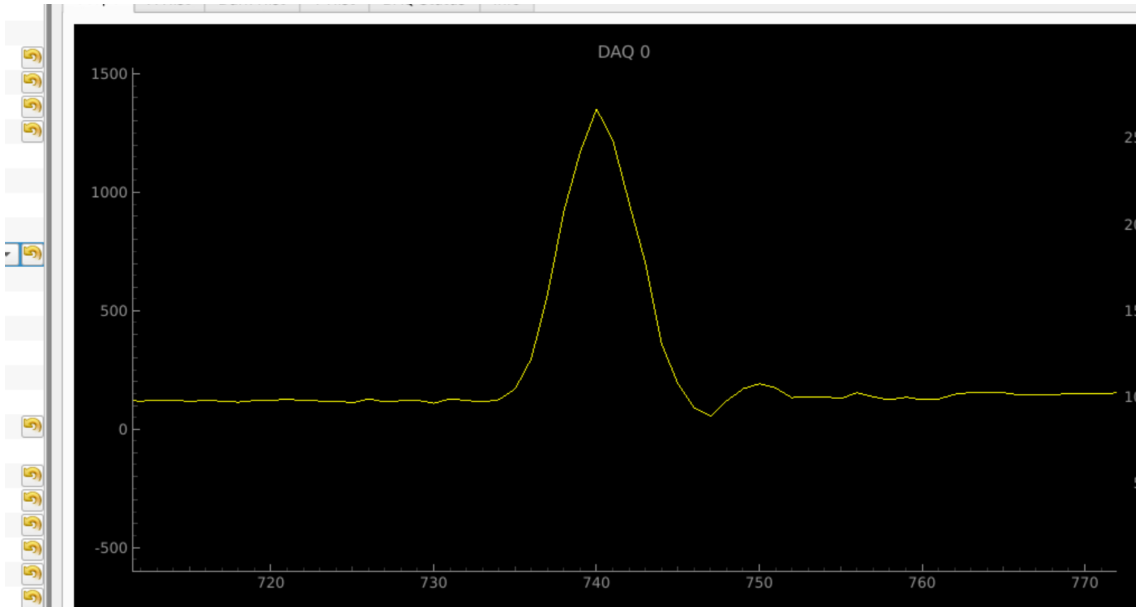
Reflections and autocorrelation and cross talk

We checked everything. Cross talk is under 0.5% (as the worst-case observed). We saw very very little in autocorrelation, except for a small cable reflection around 2%.



We applied deconvolution to Stave Mod A. saved as 00014.

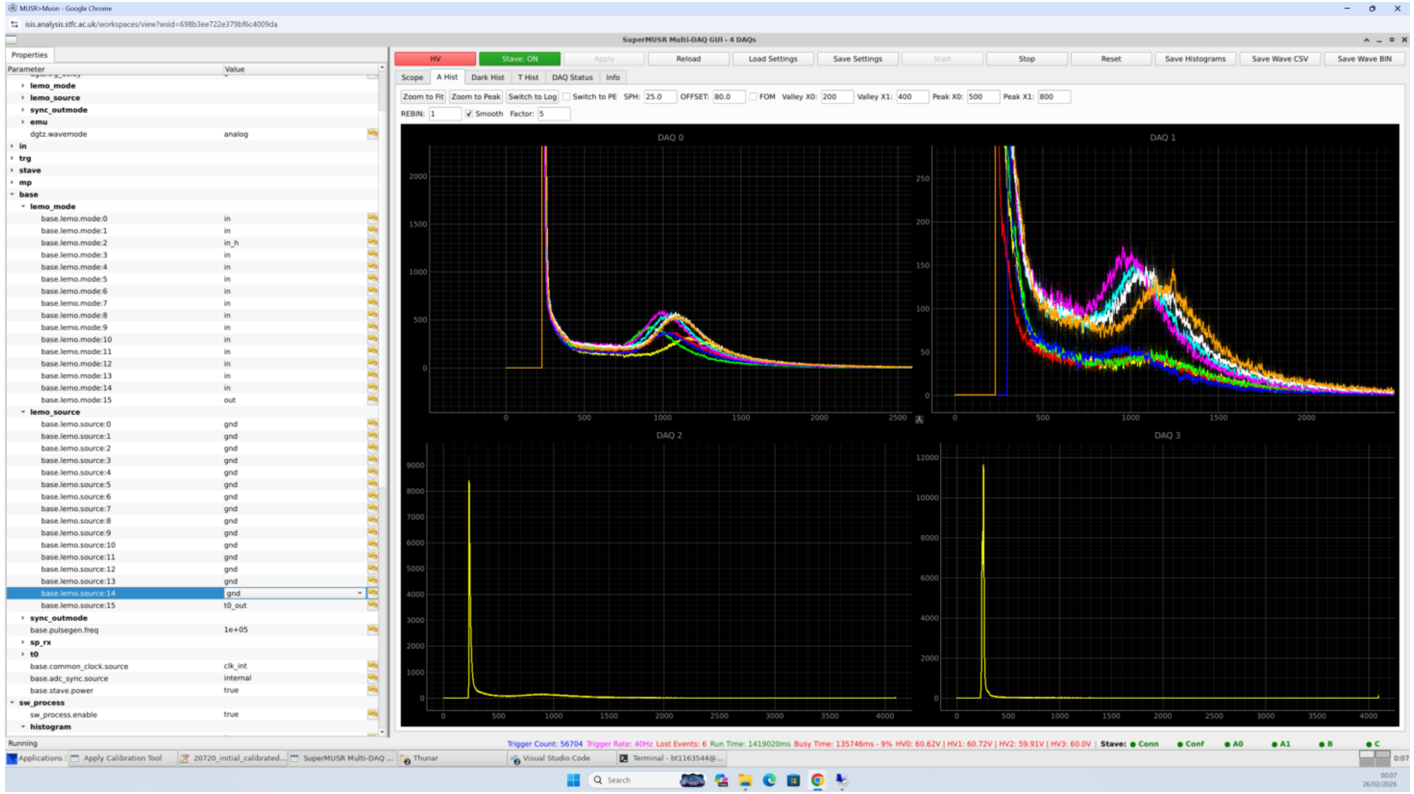




0014

<u>stv_conf_003</u>

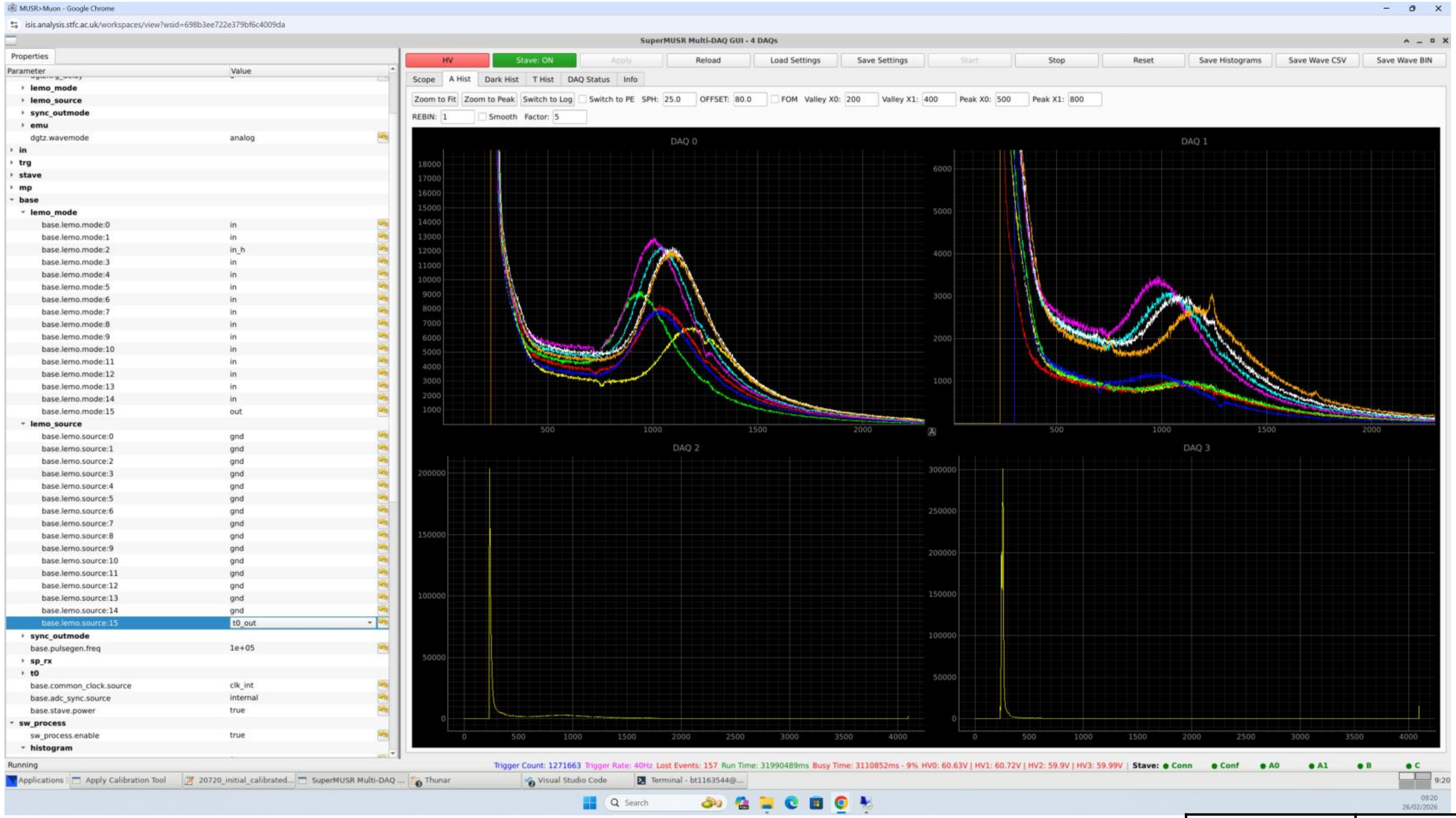
I.E two tiles were swapped over on Mod A0. Didn’t see much change. There might be optical misalignment. TO CHECK with microscope.



STAVE PID:	A0 PID:	A1 PID:	B PID:	C PID:	Table Key:
20720	20688A007	20689A006	20689B002	20692C002	stave_conf_0003

A0 PE	A1 PE	B PE	C PE
15	15	25	50

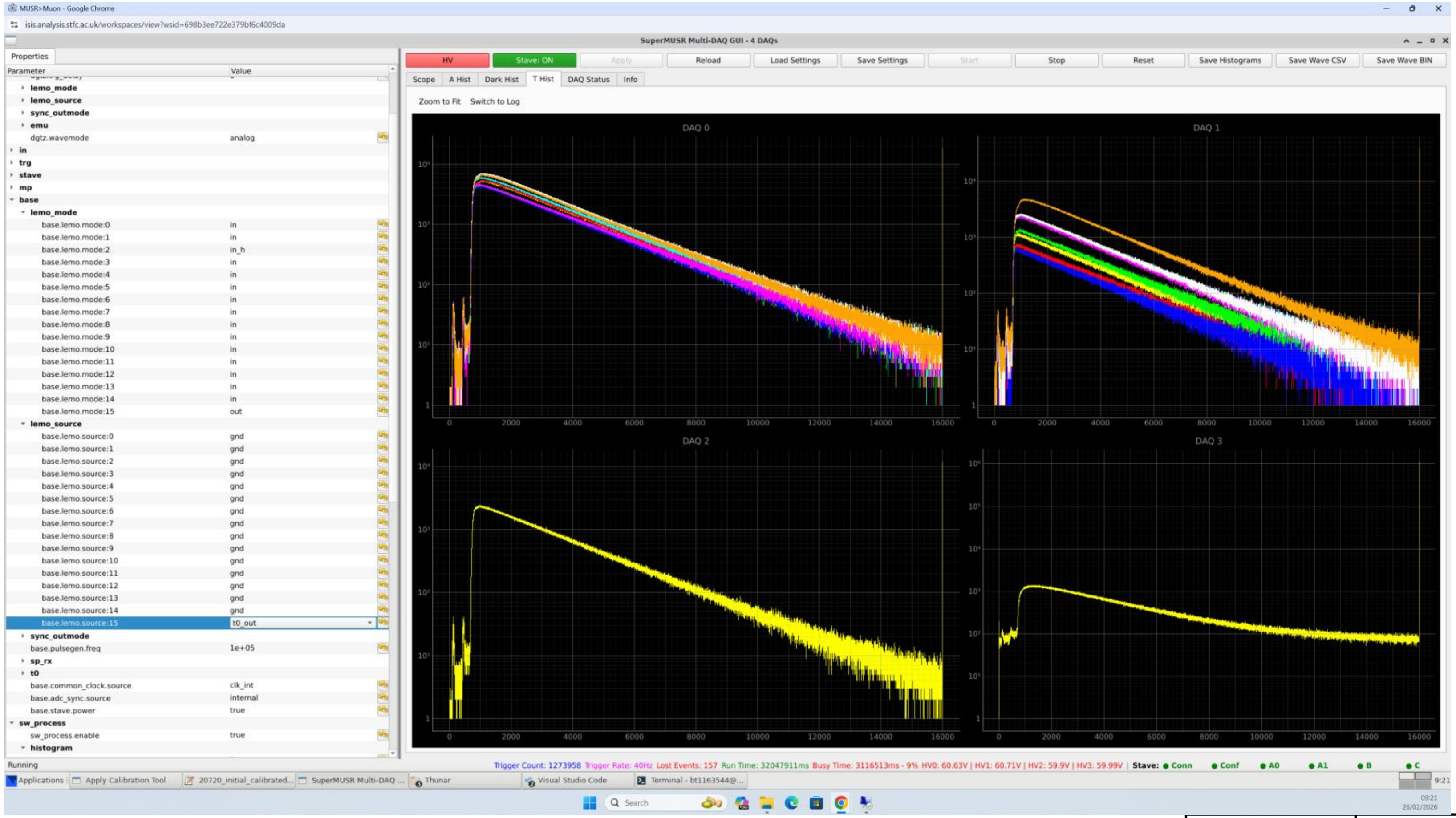
- Run 0016, long run of the same setup as tile 0014



STAVE PID:	A0 PID:	A1 PID:	B PID:	C PID:	Table Key:
20720	20688A007	20689A006	20689B002	20692C002	stave_conf_0003

A0 PE	A1 PE	B PE	C PE
15	15	25	50

- Run 0016, long run of the same setup as tile 0014



STAVE PID:	A0 PID:	A1 PID:	B PID:	C PID:	Table Key:
20720	20688A007	20689A006	20689B002	20692C002	stave_conf_0003

A0 PE	A1 PE	B PE	C PE
15	15	25	50

The protostave was connected in a permeant position, it is working on the GUI. Calibrated for Module B